Crop development and growth

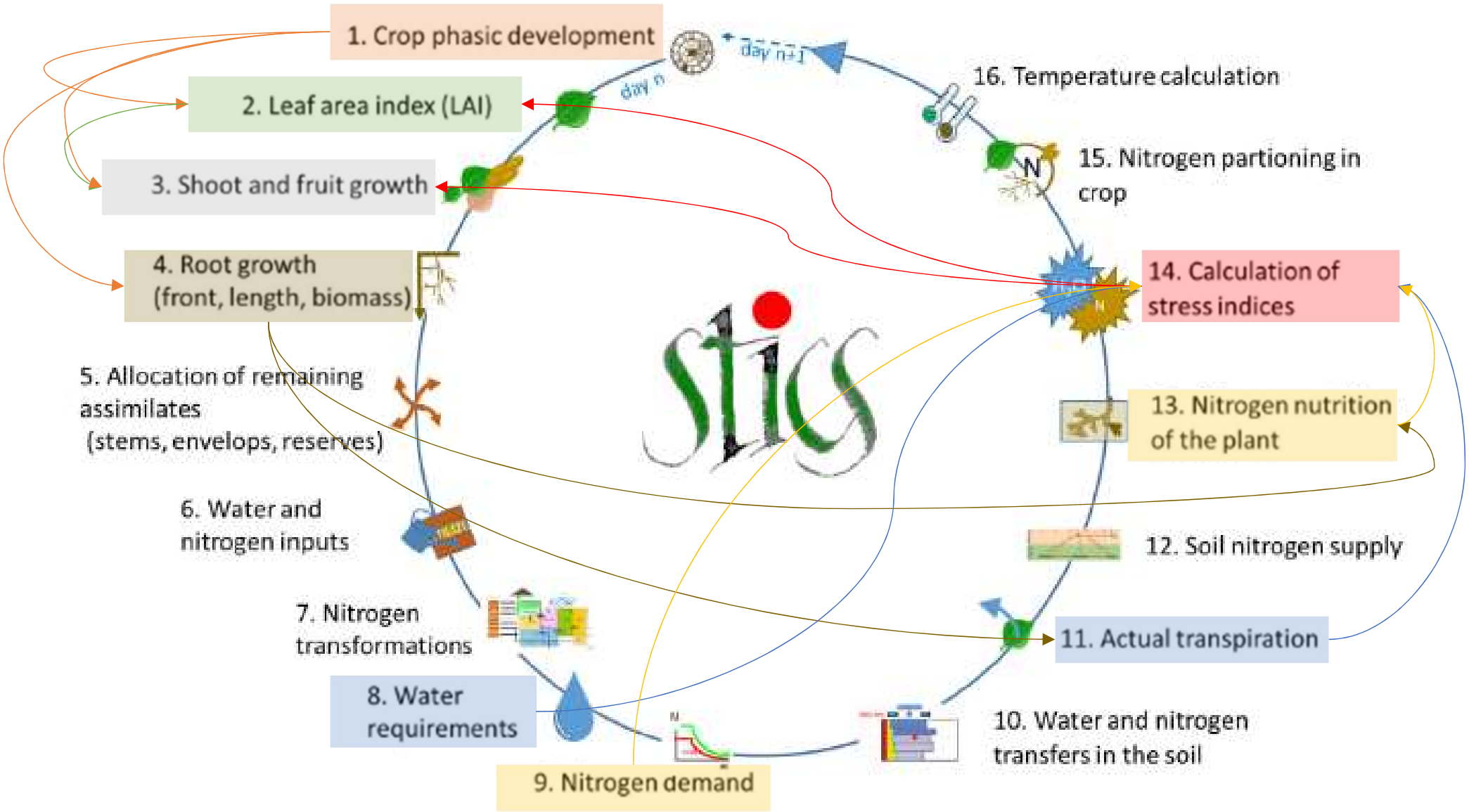


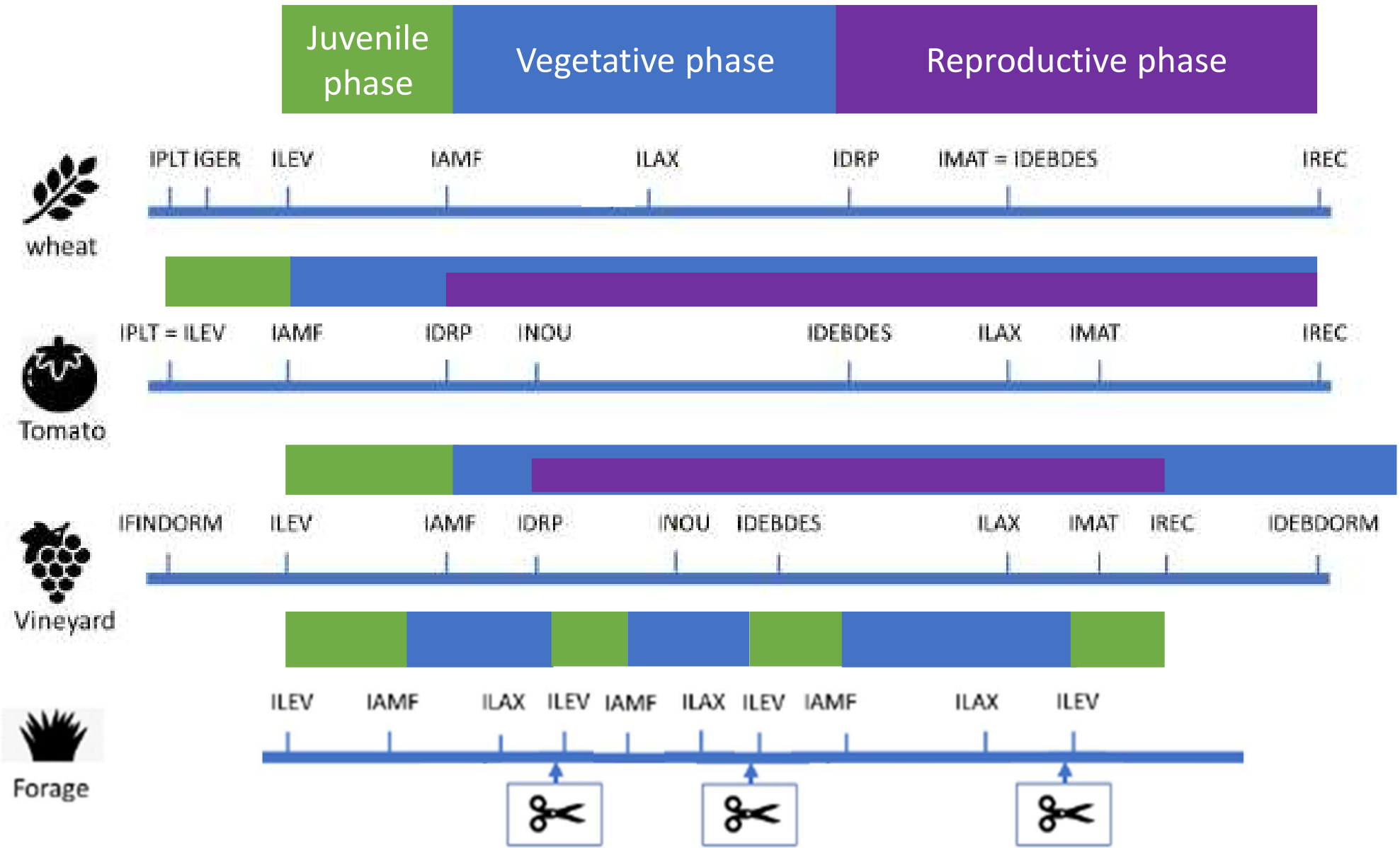
Figure 2.4. Simplified diagram of the order in STICS between processes within the daily loop.

Crop development

Crop development

Vegetative stages (Leaf Area Index)	**Reproductive stages (harvested organs filling)**
IPLT: seedling	
IGER: germination	
IDEBDORM et IFINDORM: beginning and end of dormancy	
ILEV: Emergence for annual crops or Budbreak for perennial crops	
ILET: End of plantlet sensitivity to frost	IFLO: flowering
IAMF: maximum acceleration of leaf growth, end of juvenile phase	IDRP: Onset of filling of harvested organs
ILAX: End of LAI growth, emission of new leaves stop	INOU: end of setting (indeterminate crops)
	IDEBDES: onset of water dynamics in fruits
	IMAT: physiological maturity
IREC: harvest	

Crop development

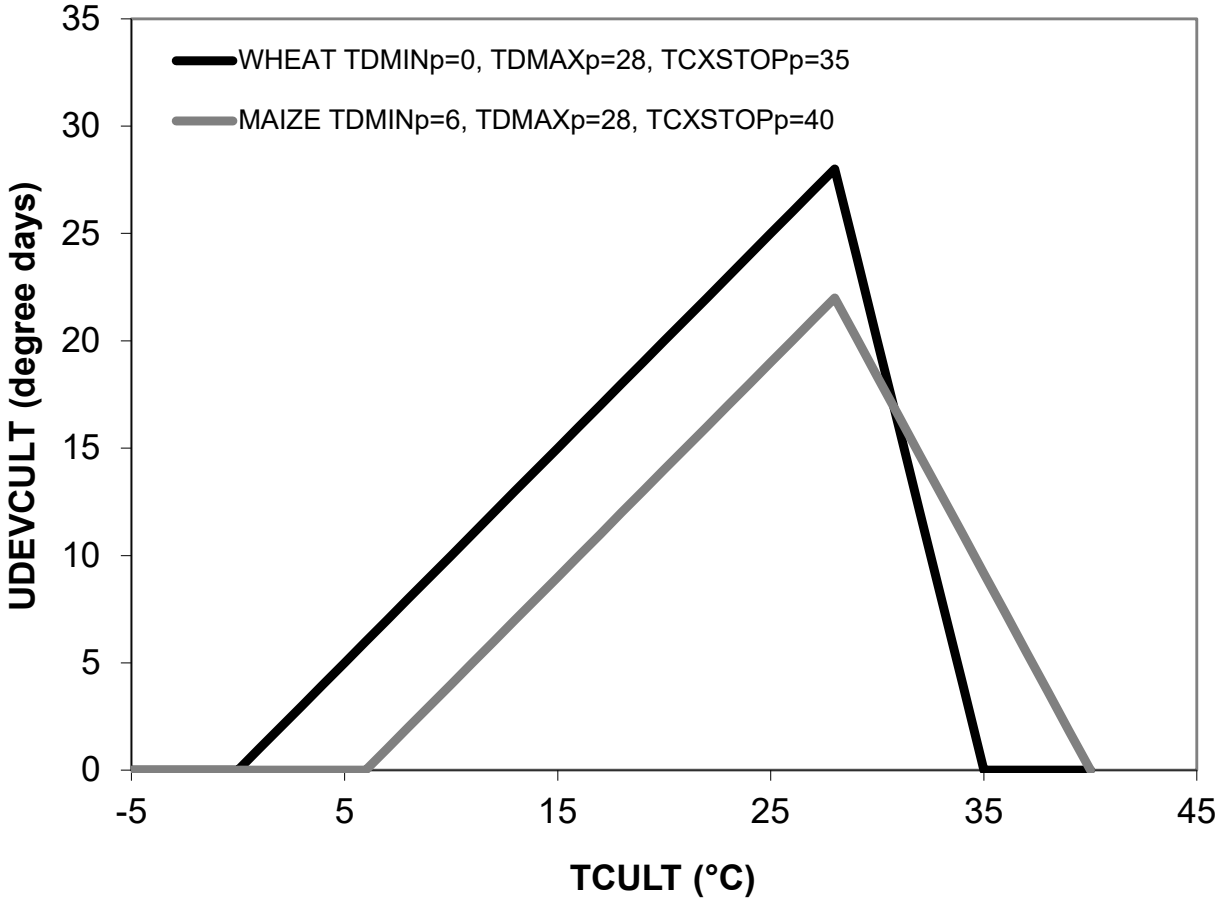


Development: a vernalo-photoperiodic-thermal time scale

Development unit of the crop (degree days) :

$$UDEVCULT(n) = Tmean(n) - TDmin_P$$

Example: Specific driving temperatures for maize and wheat



Development: a vernalo-photoperiodic-thermal time scale

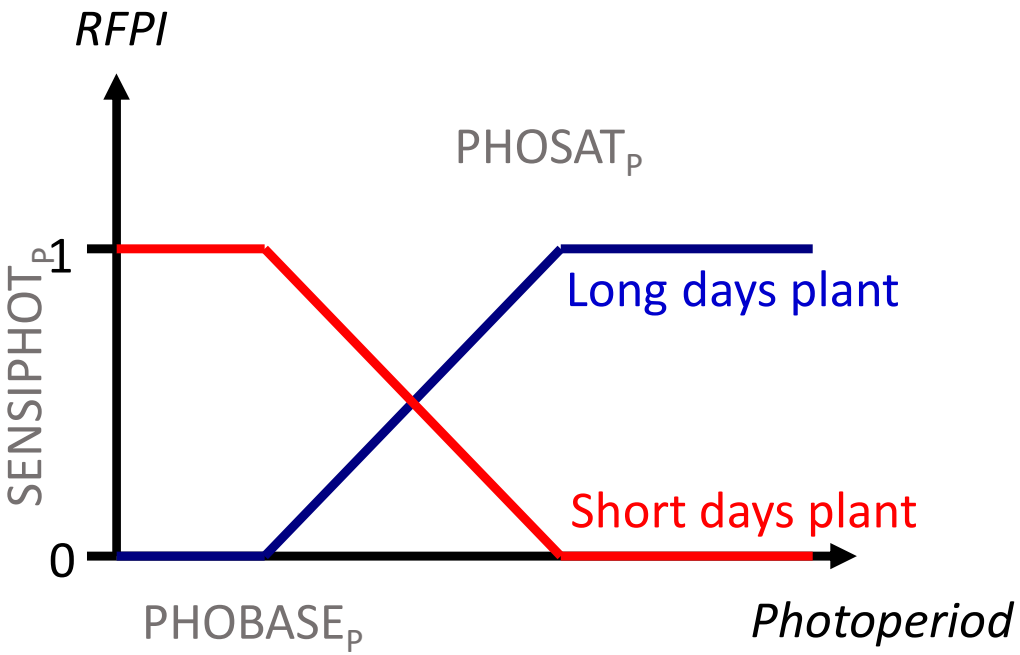
Potential development unit of the crop (degree days) :

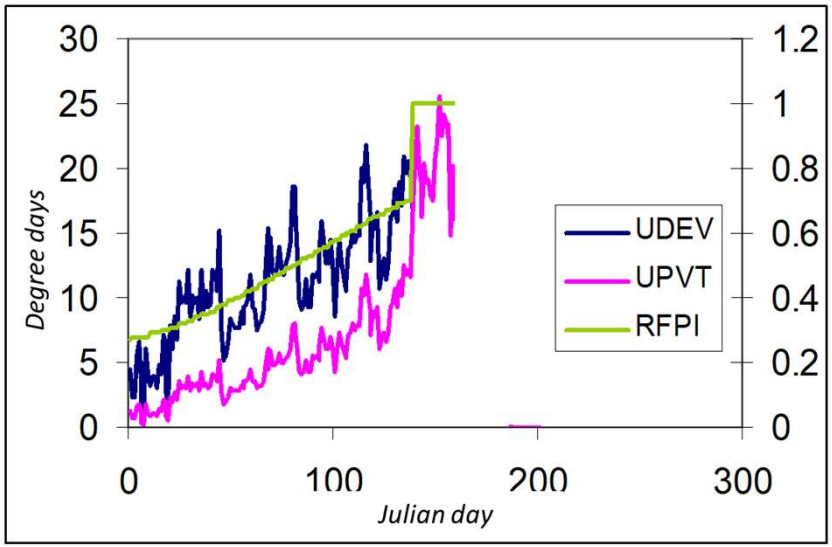
$$UDEVCULT(n) = Tmean(n) - TDmin_P$$

Actual development unit of the crop, example for a photoperiodic plant :

$$UPVT(n) = UDEVCULT(n) * RFPI(n)$$

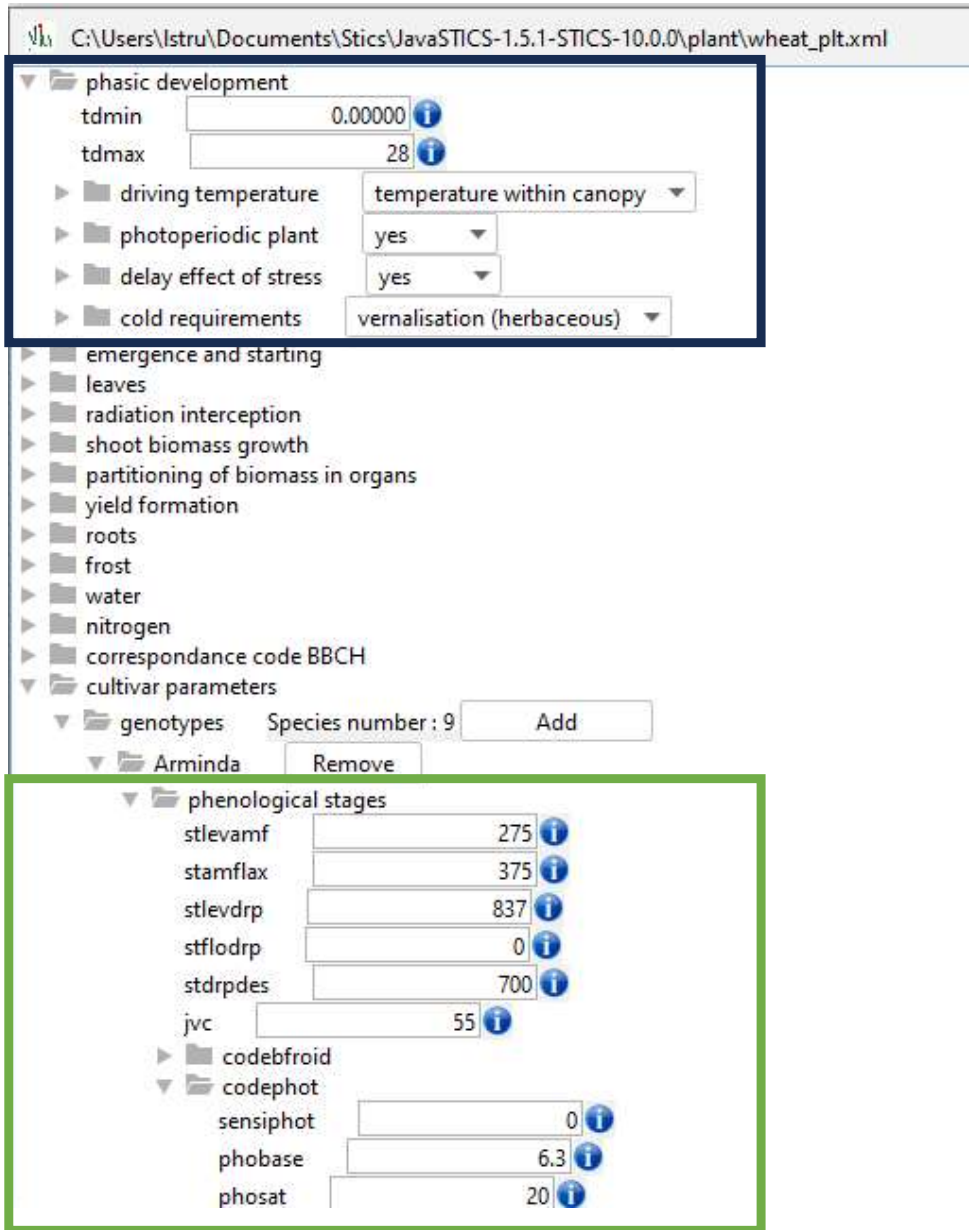
Example: *Triticum durum* (long days plant)





Development

➢ Plant parameters and options for simulation of crop development are available in the plant file:



Development: germination of annual crops

1. Moistening: yes or no

- Soil too dry, seed cannot germinate yet.
- Soil moisture ≥ moisture threshold then seed moistens.

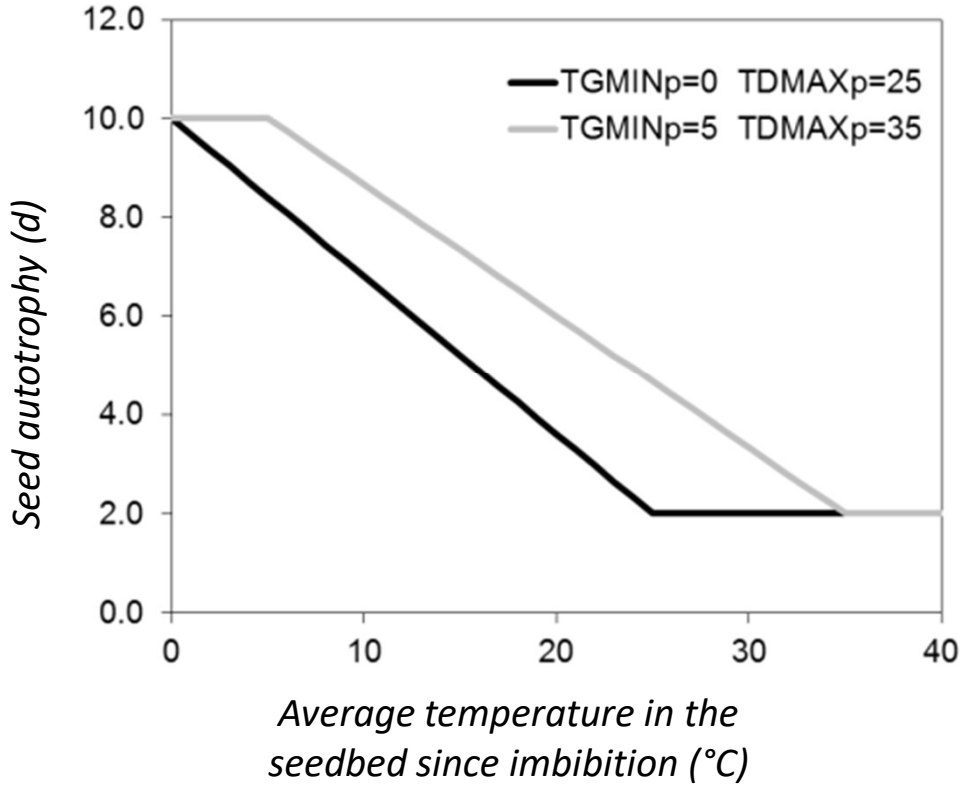
Development: germination of annual crops

1. Moistening: yes or no

- Soil too dry, seed cannot germinate yet.
- Soil moisture ≥ moisture threshold then seed moistens.

2. Maximum duration of seed autotrophy

- Autotrophy ability of seed is function of the plant and seedbed temperature



Development: germination of annual crops

1. Moistening: yes or no

- Soil too dry, seed cannot germinate yet.
- Soil moisture ≥ moisture threshold then seed moistens.

2. Maximum duration of seed autotrophy

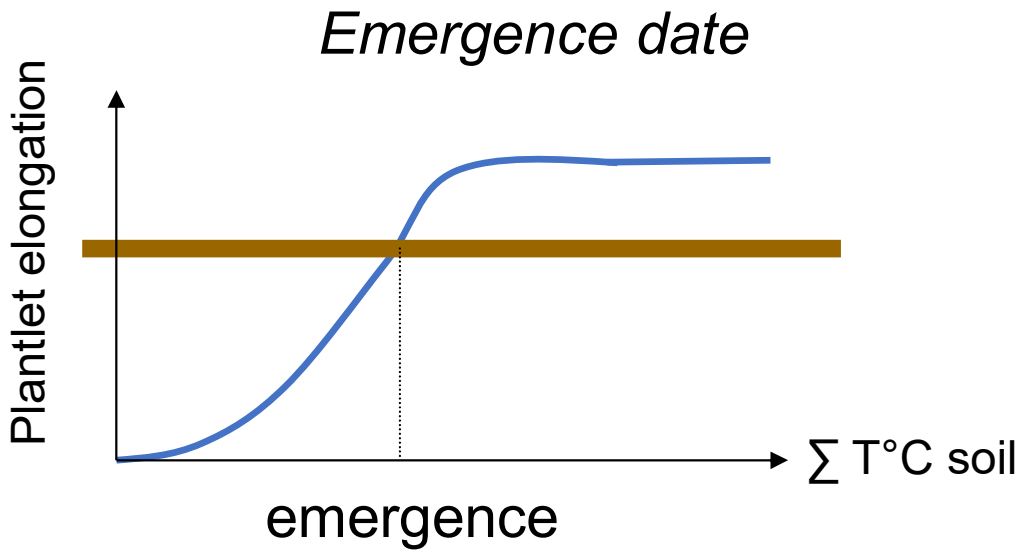
- Autotrophy ability of seed is function of the plant and seedbed temperature (respiration and consummation of resources are reduced by low temperatures)

3. Germination density

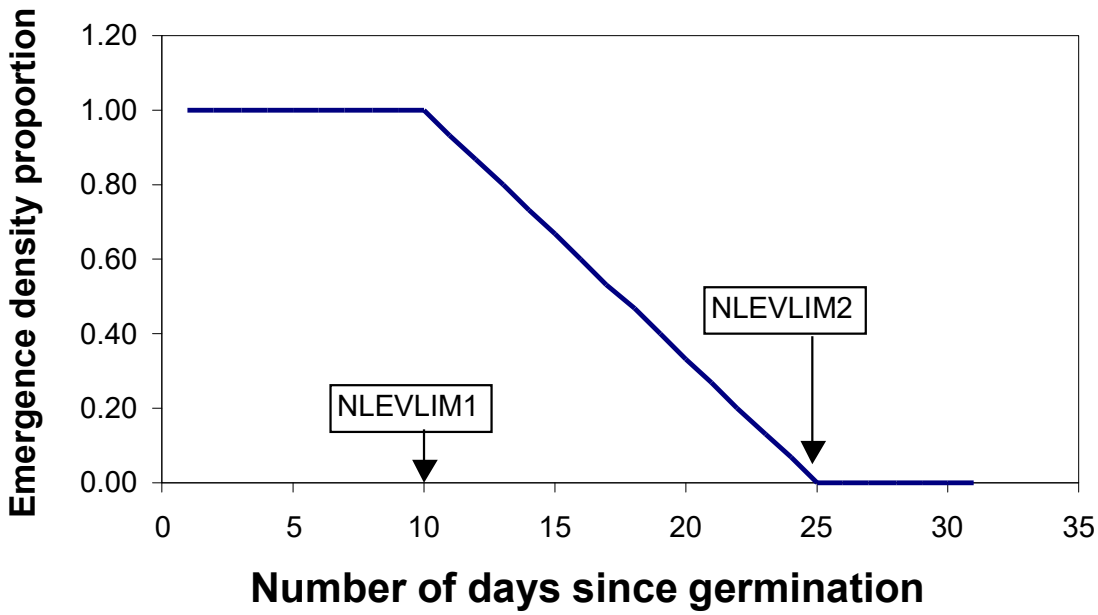
- If duration of seed germination > seed autotrophy then germination occurs but with seed loss density

Development: Emergence of annual crops

Emergence date: depending on the plantlet elongation (soil temperature response function) till it reaches the soil surface

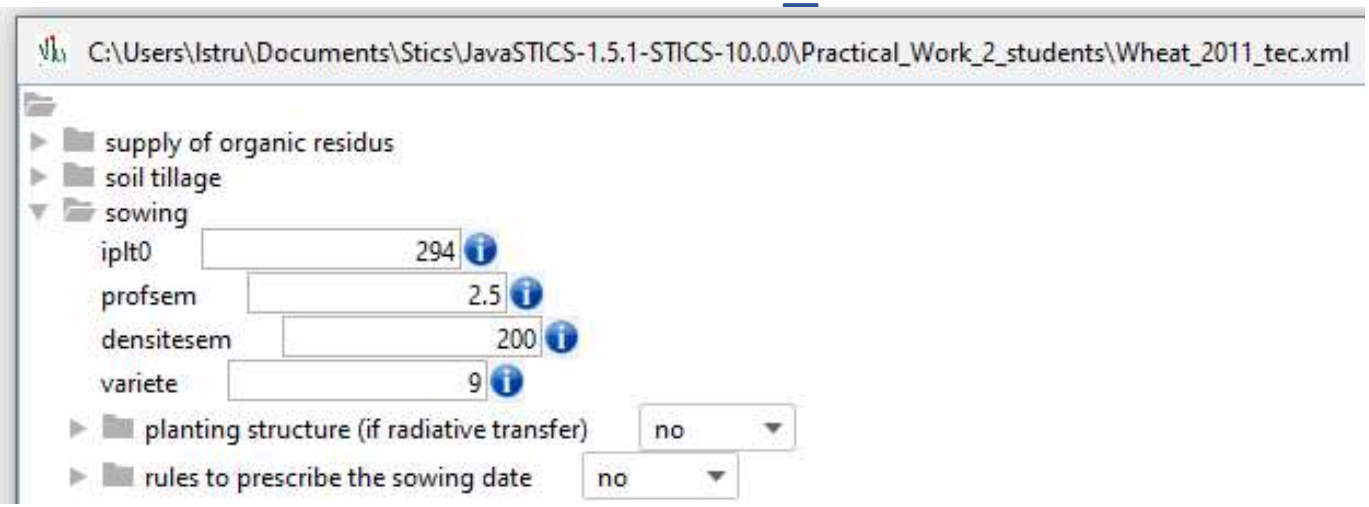


Delay in emergence (effect of cold temperatures, drought, soil crusting) => decrease of emergence density (autotrophic threshold)



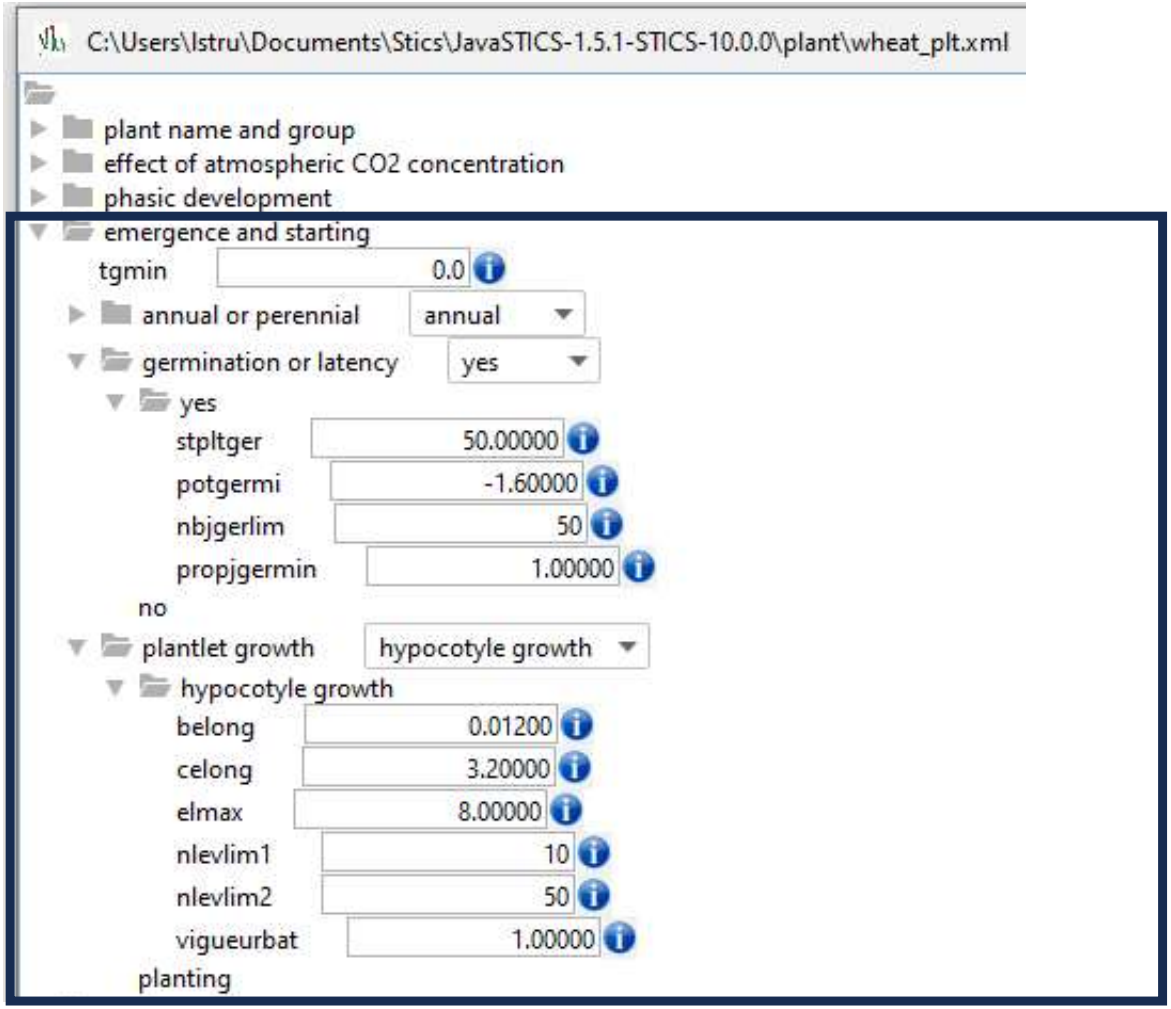
Development: Germination and emergence of annual crops

- Seedling is described in the technical file « name_tec.xml »:
  - ✓ date (julian day)
  - ✓ density (plant m^{-2})
  - ✓ depth (cm)



Development: Germination and emergence of annual crops

- Seedling is described in the technical file:
  - ✓ date (julian day)
  - ✓ density (plant m^{-2})
  - ✓ depth (cm)
- Germination and emergence depend on plant parameters:



Development: Germination and emergence of annual crops

- Seedling is described in the technical file:
  - ✓ date (julian day)
  - ✓ density (plant m^{-2})
  - ✓ depth (cm)
- Germination and emergence depend on plant parameters described in the plant file
- Seed moistening depends on soil humidity
- Germination and emergence of the crop is driven by soil temperature and can be delayed by:
  - ✓ cold temperature
  - ✓ soil droughtness
  - ✓ soil crusting
- The model simulates a final crop density depending on seedling density and eventual losses during germination and emergence phases

Do you have any question ?

Root system

Root system: root front growth

- Potential root front growth is function of soil temperature and root apex elongation rate (cultivar parameter: cm °Cd^{-1}):

  - ✓ Root front growth can be slowed down by:
    - Soil dryness (soil moisture < soil moisture at wilting point)
    - Soil bulk density
    - Anoxia (soil moisture > soil moisture at field capacity)

Root system: root density

- Standard profile: root growth and distribution in soil depends on plant parameters

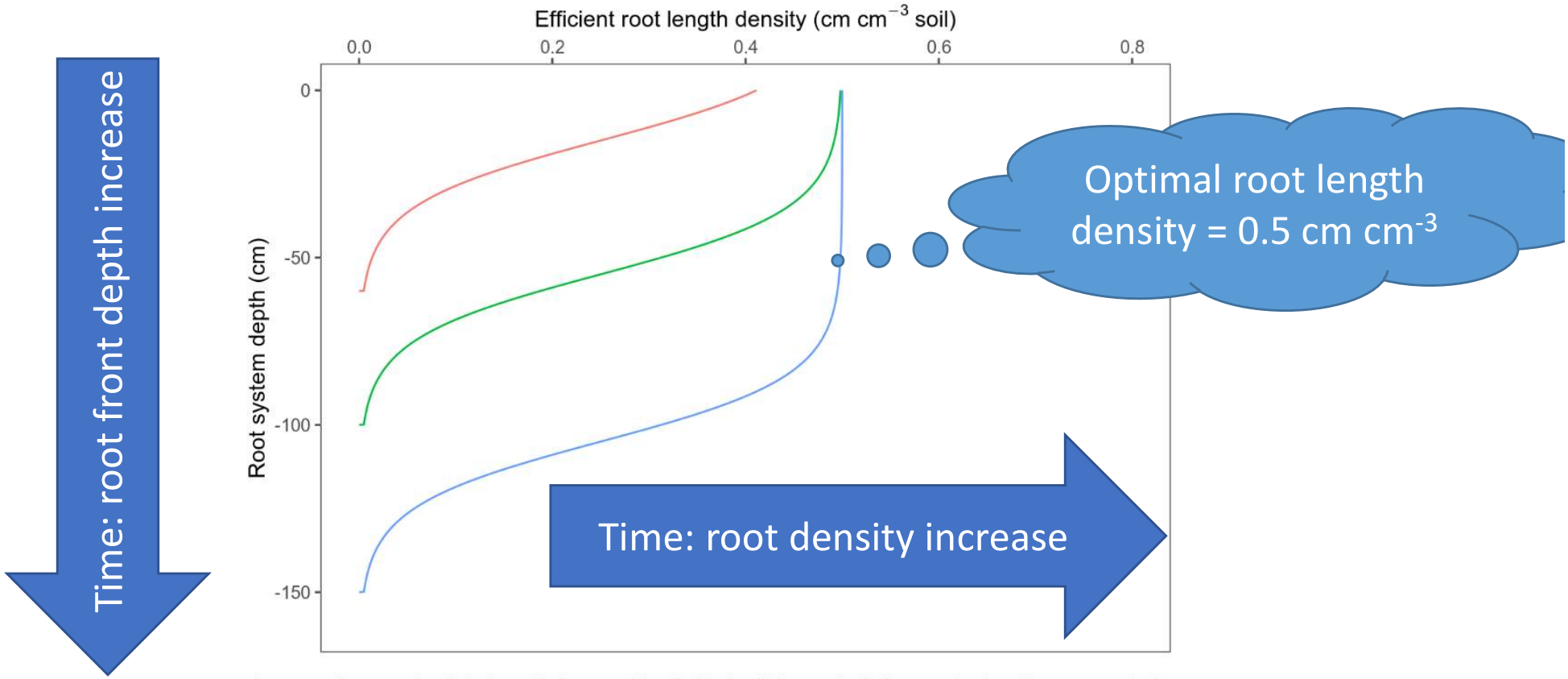


Figure 5.4. Reference root length density profile for rapeseed described by the efficient root density *lracz* as a function of the root system depth *z* and according to the root front depth *zrac*.

Root system: root density

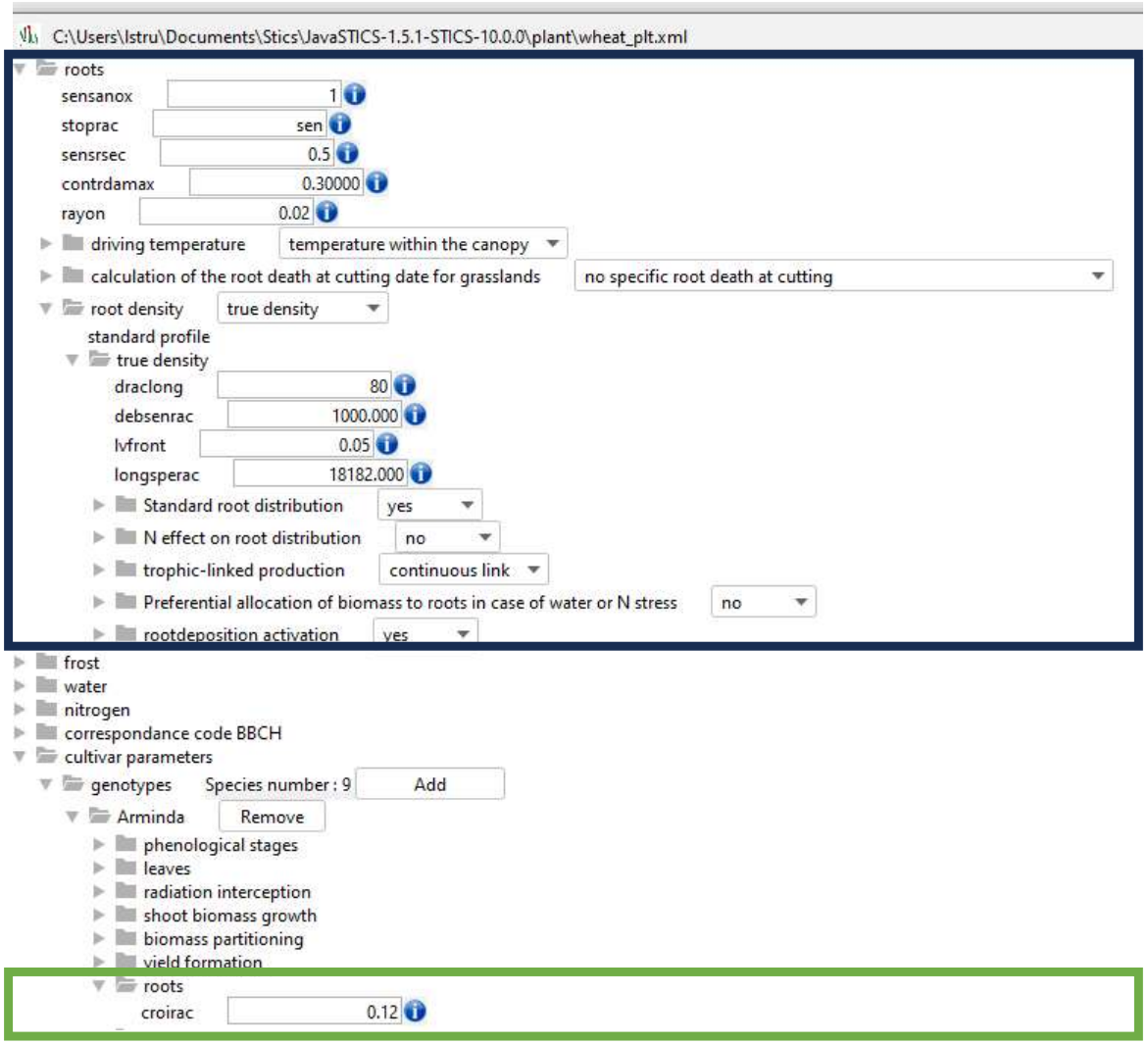
- True density: root growth and distribution in soil depends on plant parameters and soil environment:

  - ✓ Root growth can be slowed down by:
    - Soil dryness (soil moisture < soil moisture at wilting point)
    - Soil bulk density
    - Anoxia (soil moisture > soil moisture at field capacity)
    - Crop biomass production (trophic link)

  - ✓ Root distribution can be influenced by soil N availability

Root system:

- Plant parameters and options for simulation of the root system are available in the plant file:



- Do you have any question ?

Abiotic stresses

Calculation of water balance

- Soil evaporation and crop transpiration demand in function of the leaf area index (crop coefficient approach)

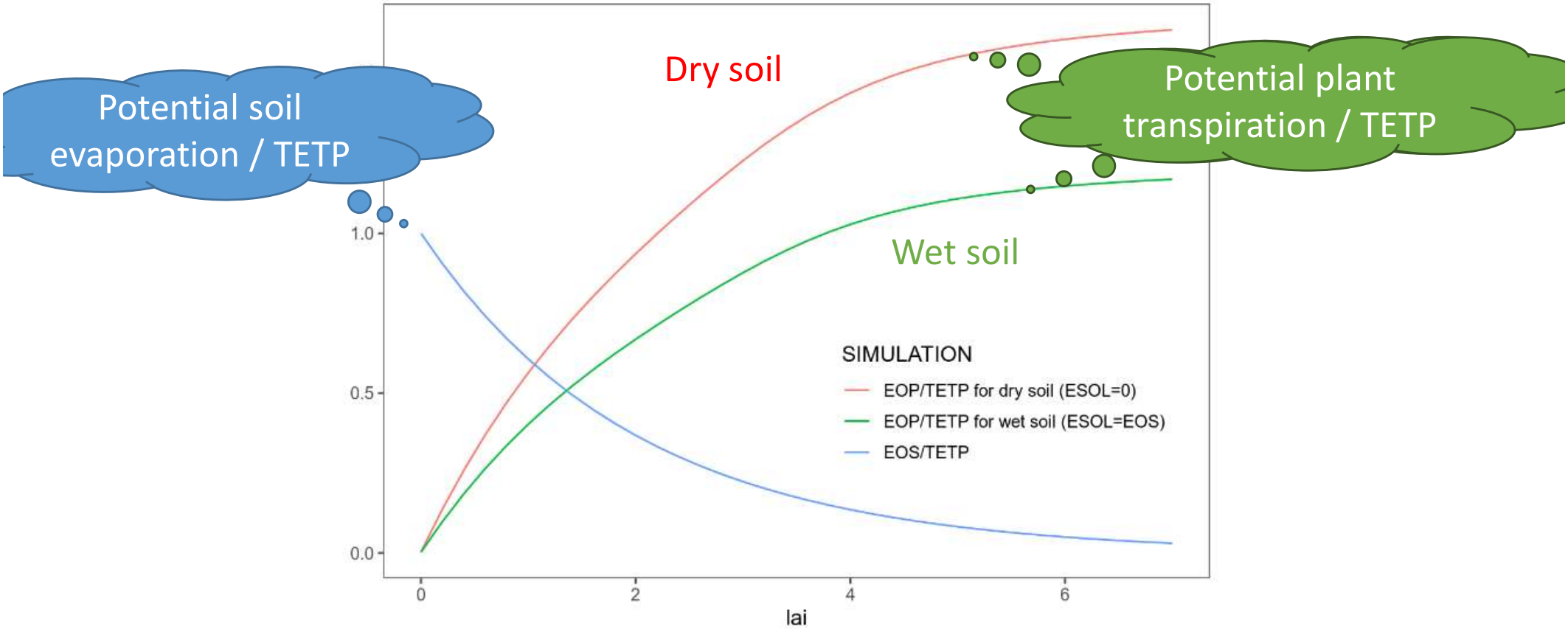
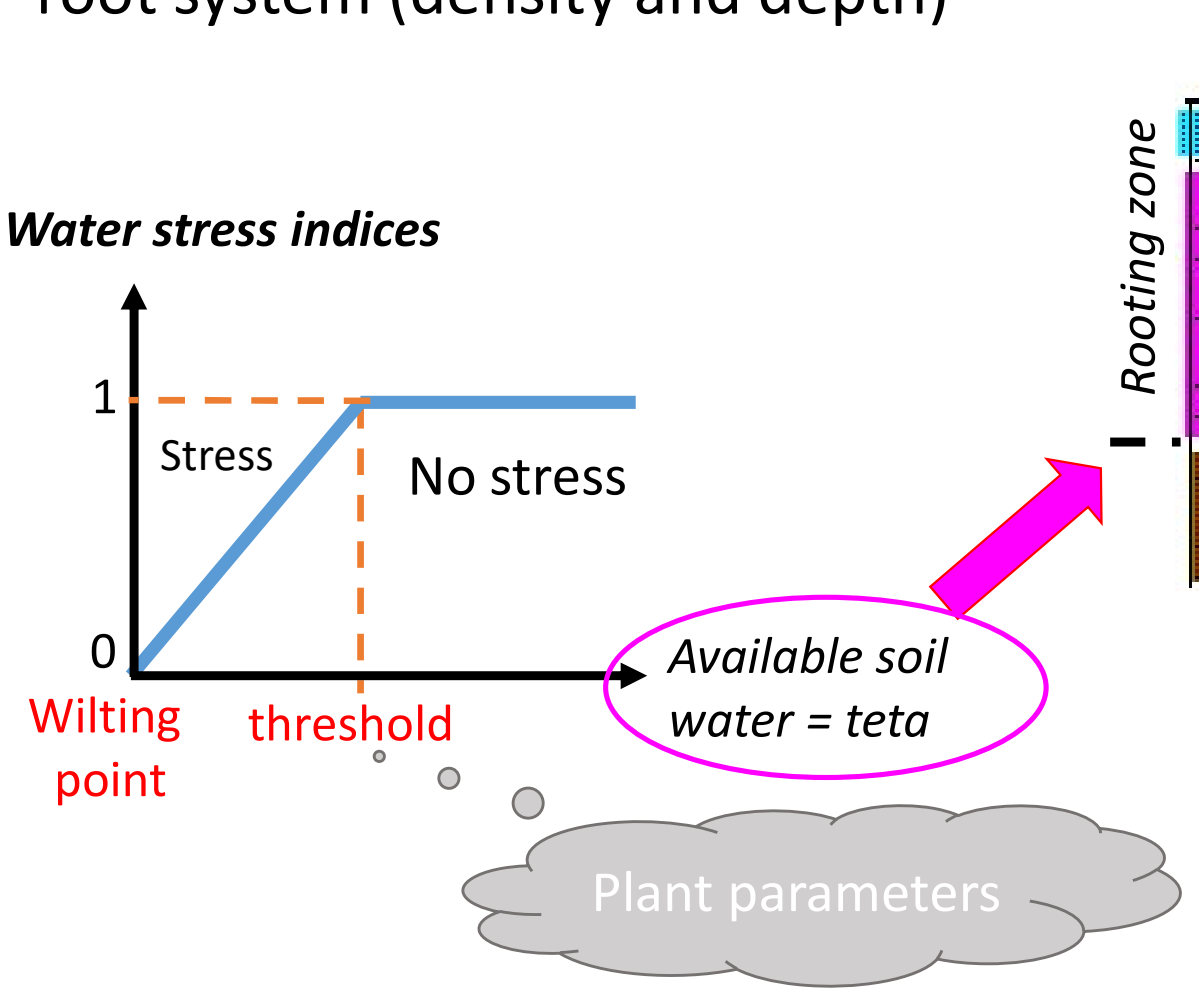
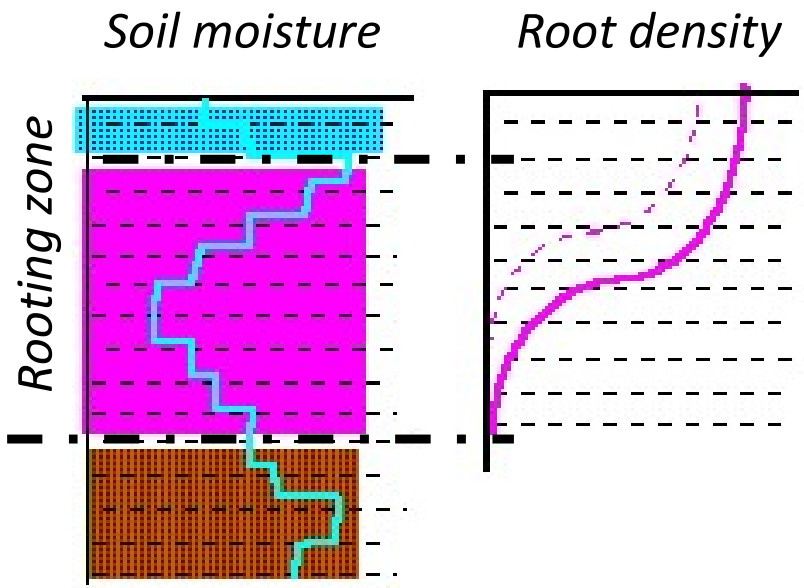


Figure 11.7. Relative evaporative demand applied to soil ($eos/tetp(t)$) and crop ($eop/tetp(t)$) accounting for the actual soil surface water state ($esol = eos$ or $esol = 0$) for canopy qualified in lai.

Root system and calculation of water stress

- Crop actual transpiration depends on soil water content and root system (density and depth)

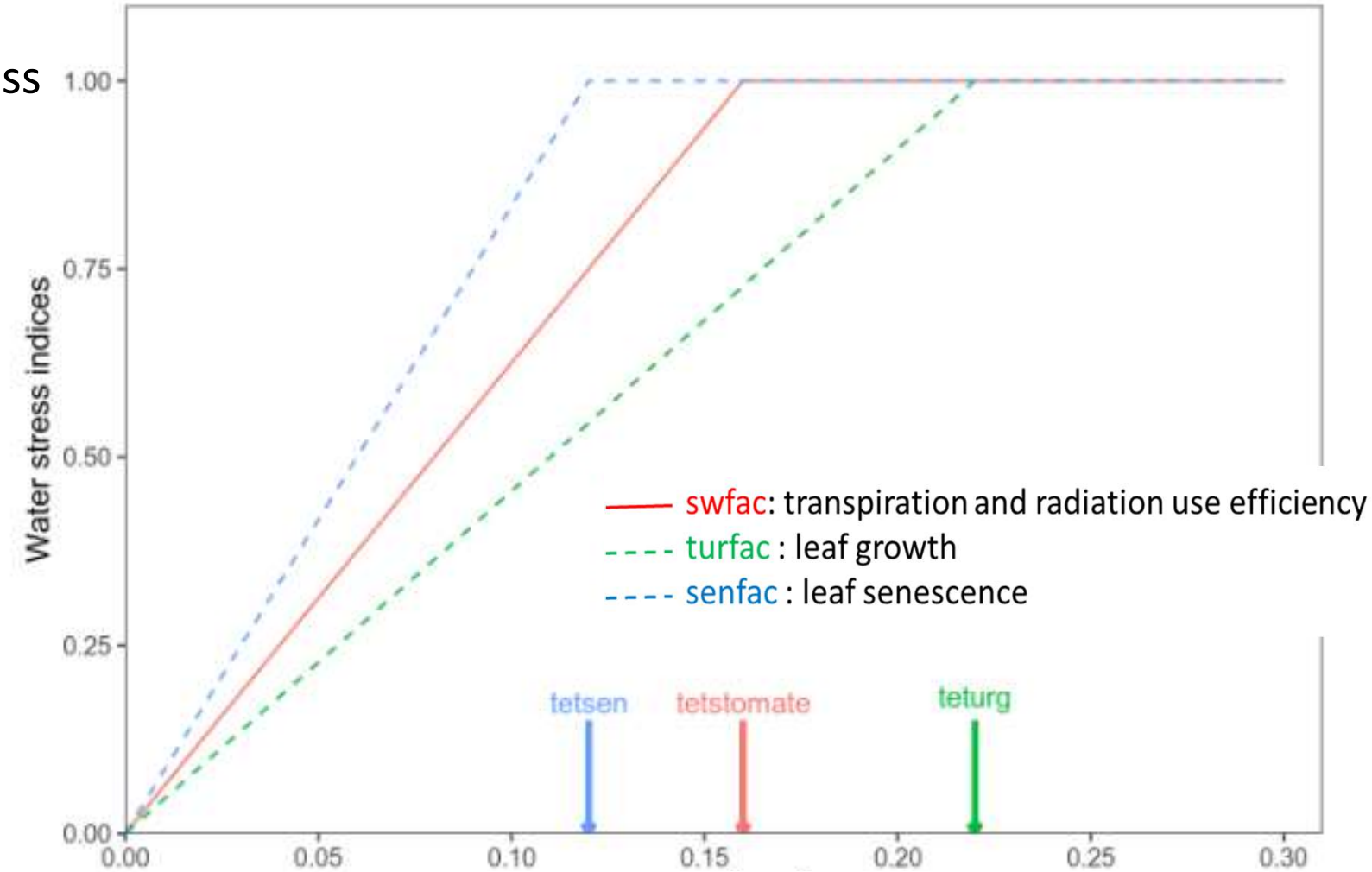




Water stress indices

- There is 3 water stress indices :

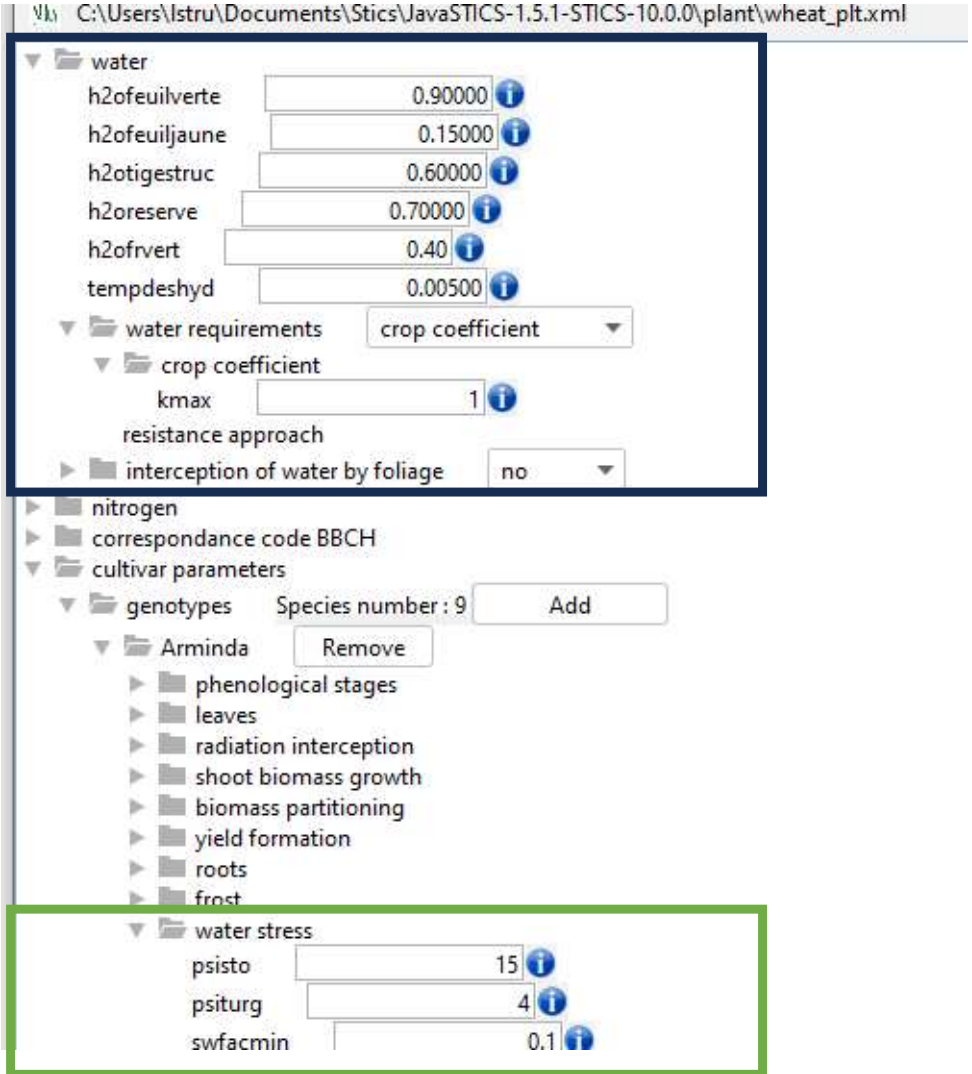
No stress



Available soil water = teta (m^3 m^{-3})

Water stress

- Plant parameters and options for simulation of water consumption and water stress are available in the plant file:



Water stress

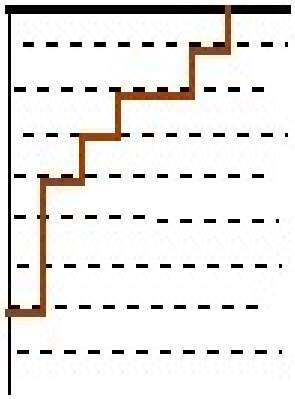
- Plant parameters and options for simulation of water consumption and water stress are available in the plant file
- Water stress appears when soil water availability in the rooting depth do not satisfy the crop transpiration demands
- There is two options for simulation of crop water demands:
  - ✓ coefficient approach for homogeneous ground cover
  - ✓ resistive approach for row crops or inter-crop
- There is 3 water stress indices impacting different processes:
  1. TURFAC : leaf growth
  2. SWFAC : transpiration and RUE
  3. SENFAC : leaf senescence

- Do you have any question ?

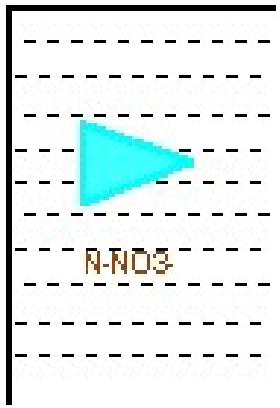
Root system and calculation of nitrogen stress

N available in the soil

$[NO_3]$

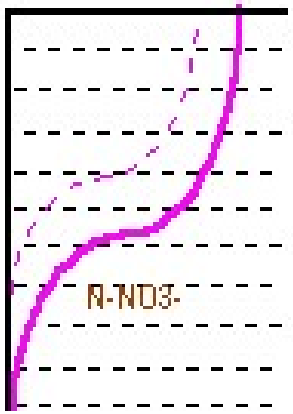


$[NO_3]$ fluxes

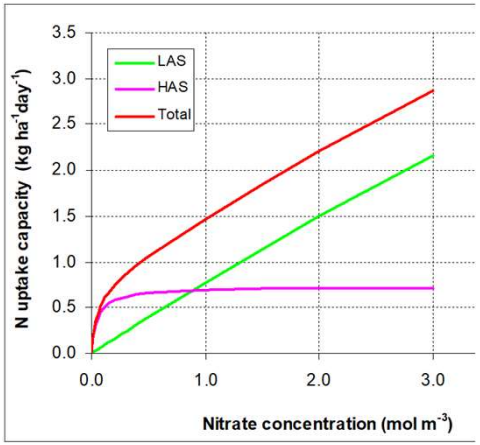


N uptake capacity

Root density

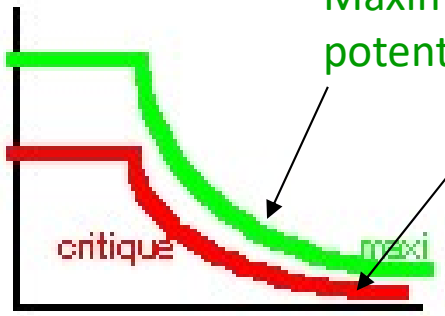


Absorption



N supply = min(N available in soil, N uptake capacity)

[N]



MS

N demand

N uptake =

min (N demand, N supply)

N fixation by leguminous

Potential N fixation
f(species and phenology)

Effect of
temperature

Effect of soil moisture

$$fixreel(i) = fixpot(i) \cdot fxt(i) \cdot fxa(i) \cdot \min\left(fxw(i) \cdot fxn(i)\right)$$

Effect of anoxia

Effect of soil N content

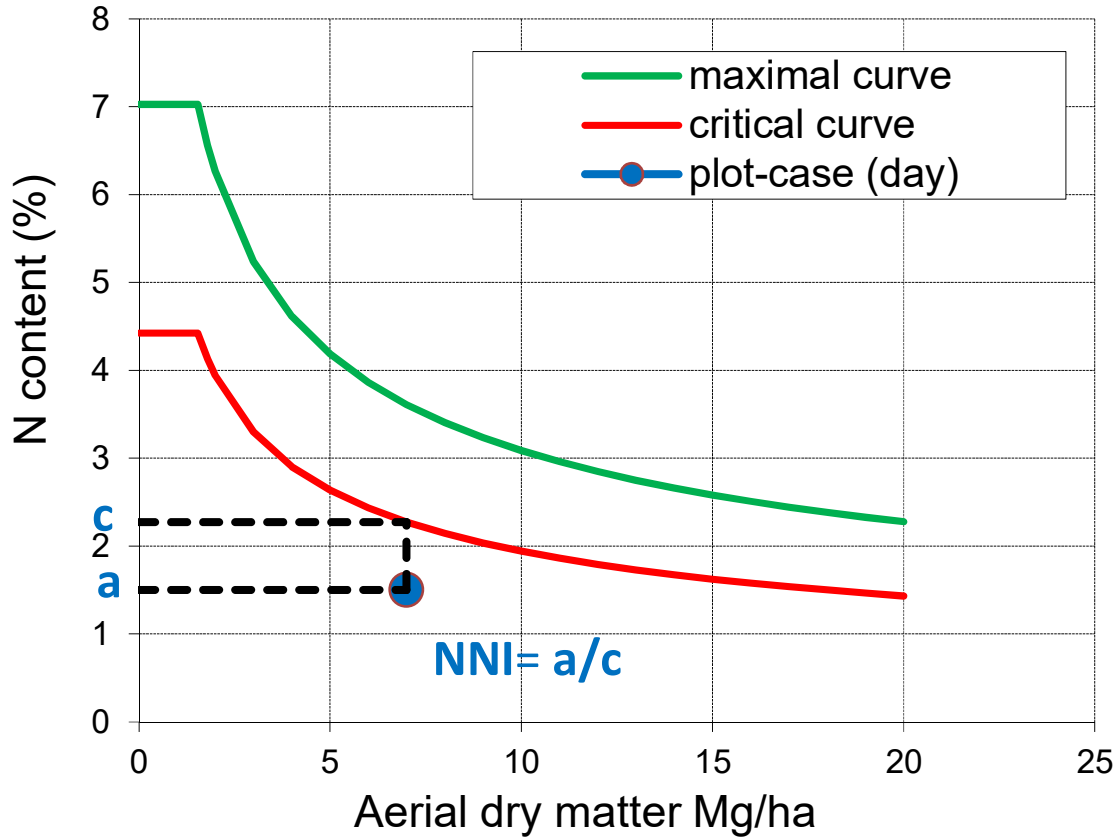
Calculation of nitrogen stress

- Nitrogen Nutrition Index (NNI):

N demand
f(species)

N requirement
f(species)

Actual N absorption

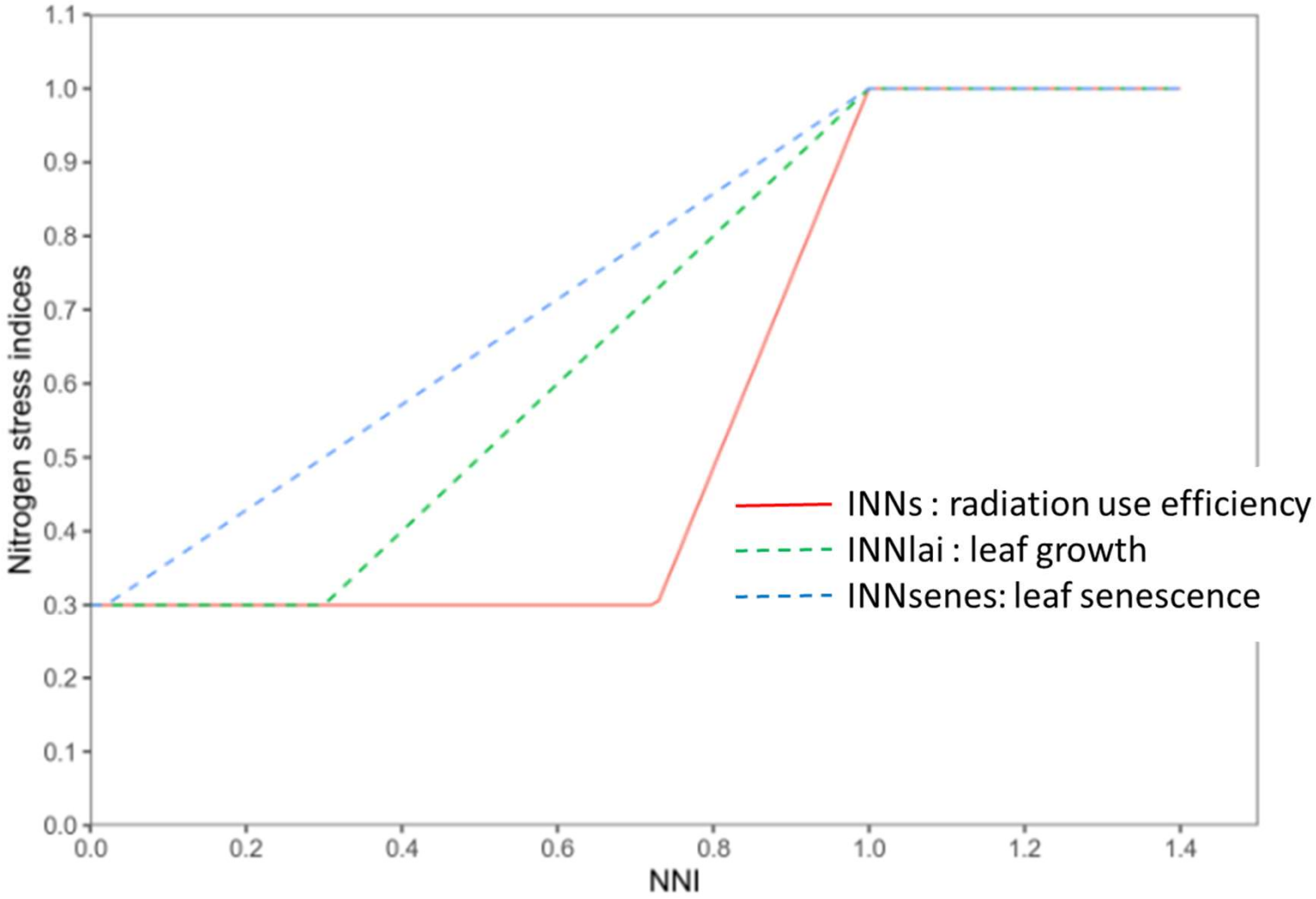


$[N]_c$ = minimal N content to have maximal biomass production

Nitrogen Nutrition Index (NNI) : NNI (d) = a/c

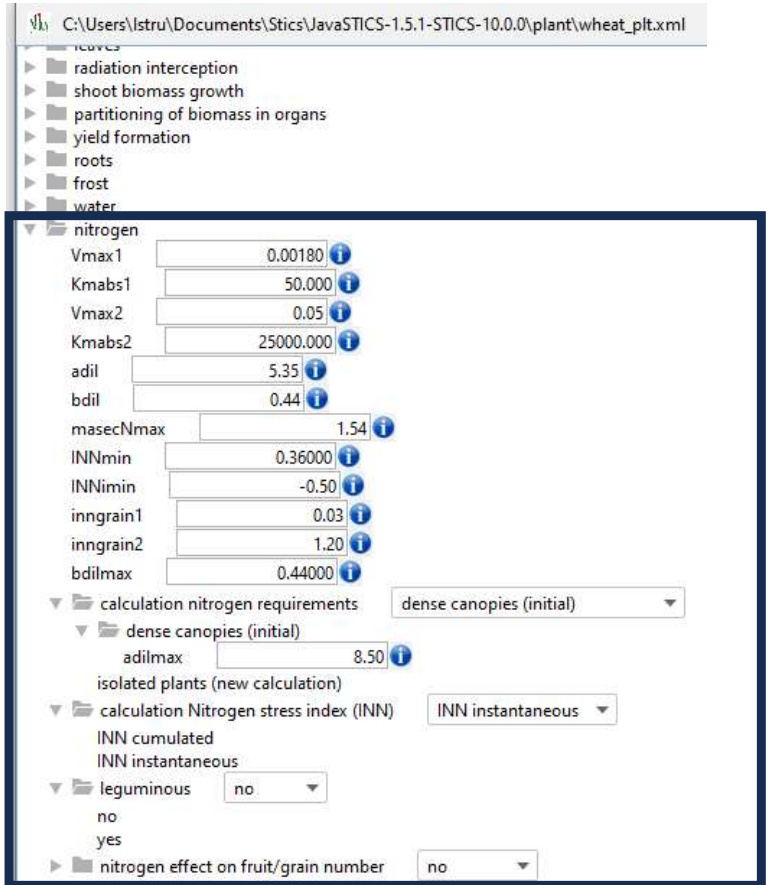
Nitrogen stress indices

- There is 3 nitrogen stress indices :



Nitrogen stress

- Actual N uptake by the crop is equal to the minimum between crop N demand and N supply.
- Plant parameters and options for simulation of N demand, N uptake and N stress are available in the plant file

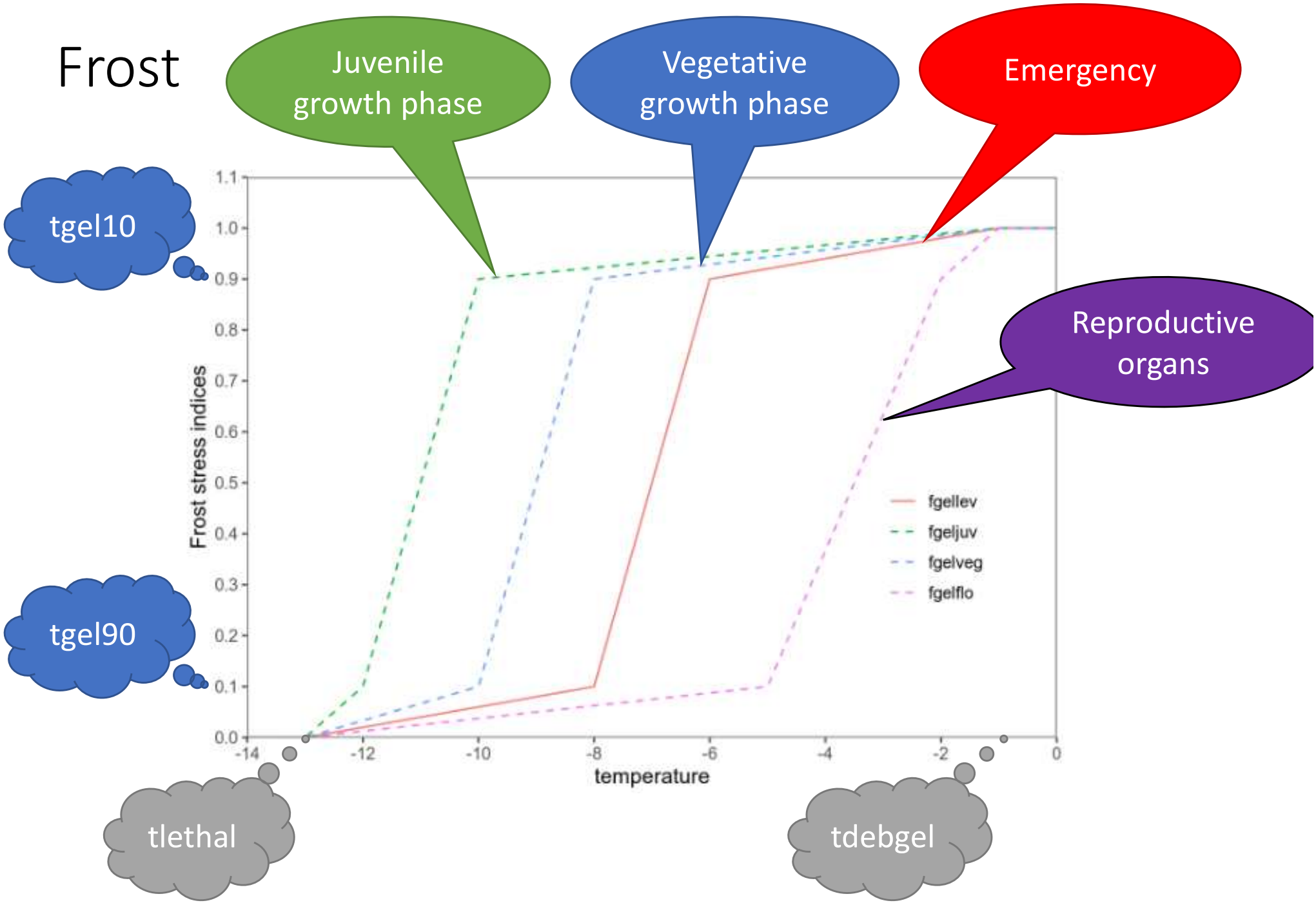


Nitrogen stress

- Actual N uptake by the crop is equal to the minimum between crop N demand and N supply.
- Plant parameters and options for simulation of N demand, N uptake capacity and N stress are available in the plant file
- N stress depends on the Nitrogen Nutrition Index of the crop
- There is 3 N stress indices impacting different processes:
  1. or 2. INNLAI : leaf growth
  2. or 1. INNs : RUE
  3. INNsenes : leaf senescence

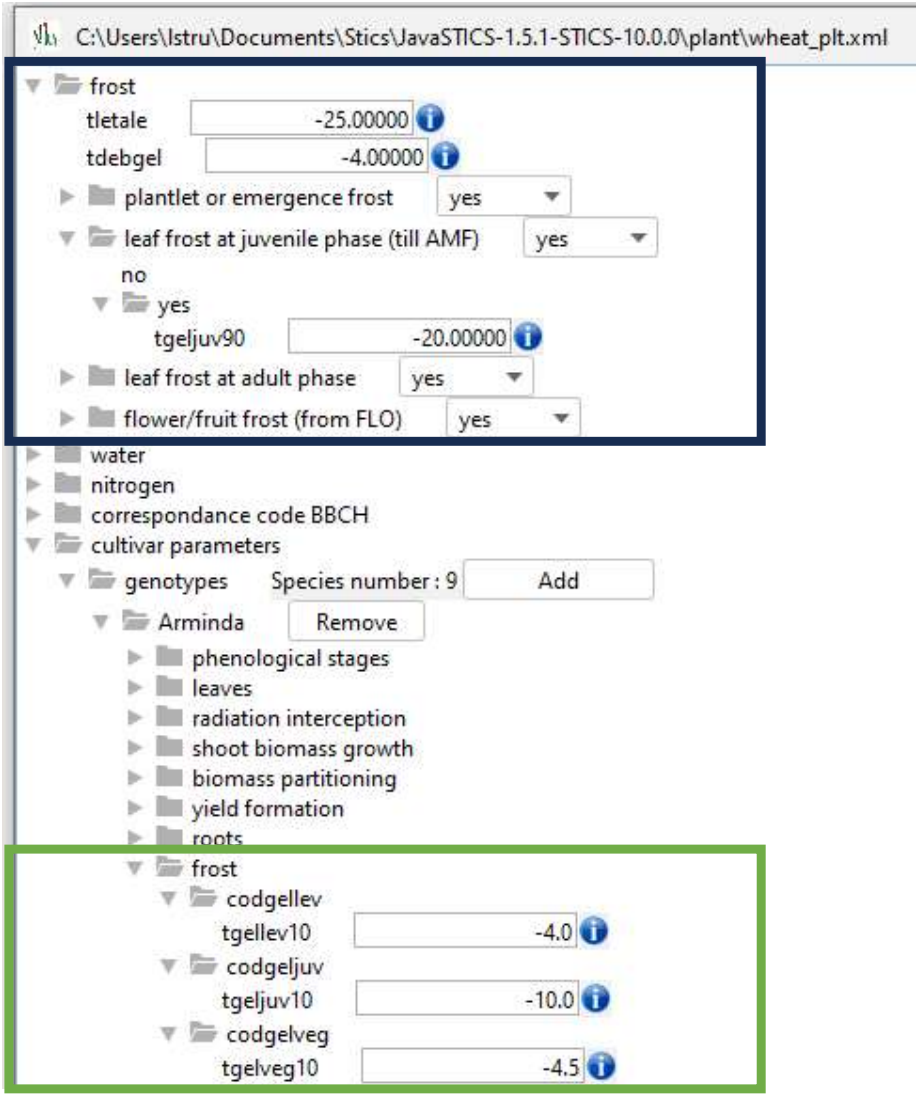
- Do you have any question ?

Frost


Juvenile growth phase
Vegetative growth phase
Emergency
Reproductive organs
tgel10
tgel90
tlethal
tdebgel
Frost stress indices
temperature
fgellev
fgeljuv
fgelveg
fgelflo
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
-14
-12
-10
-8
-6
-4
-2
0

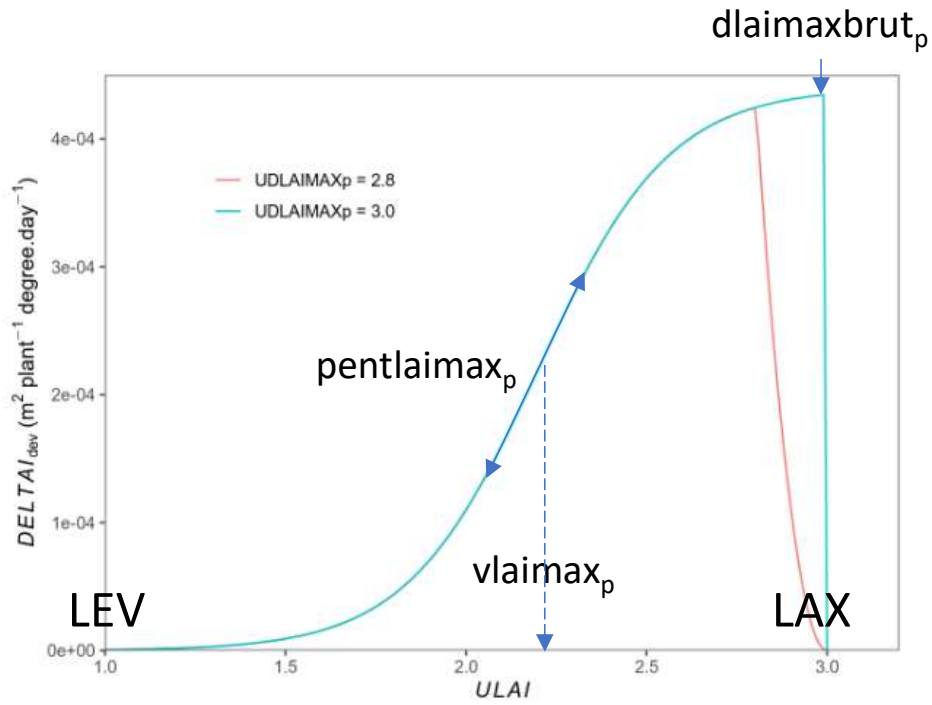
Frost

- Plant parameters are available in the plant file



Leaf area index
and
radiations interception

Leaf area index growth



Normalized development unit of LAI

Potential LAI growth (m^2 plant^{-1} dd^{-1}):

Potential growth

$$DELTAI_{dev}(I) = \frac{DLAIMAXBRUT_P}{1+\exp\left[PENTLAIMAX_P \cdot \left(VLAIMAX_P - ULAI(I)\right)\right]}$$

Actual growth (m^{2}m^{-2} dd^{-1}):

Effect of density

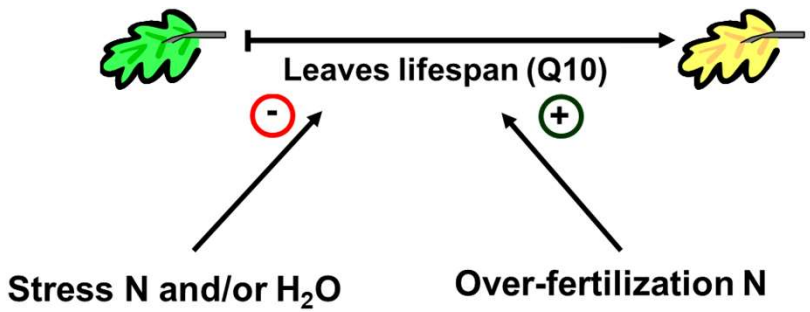
$$DELTAI_1(I) = DELTAI_{dev}(I) \cdot DELTAI_T(I) \cdot DELTAI_{dens} \cdot DELTAI_{stress}(I)$$

Effect of abiotic stresses

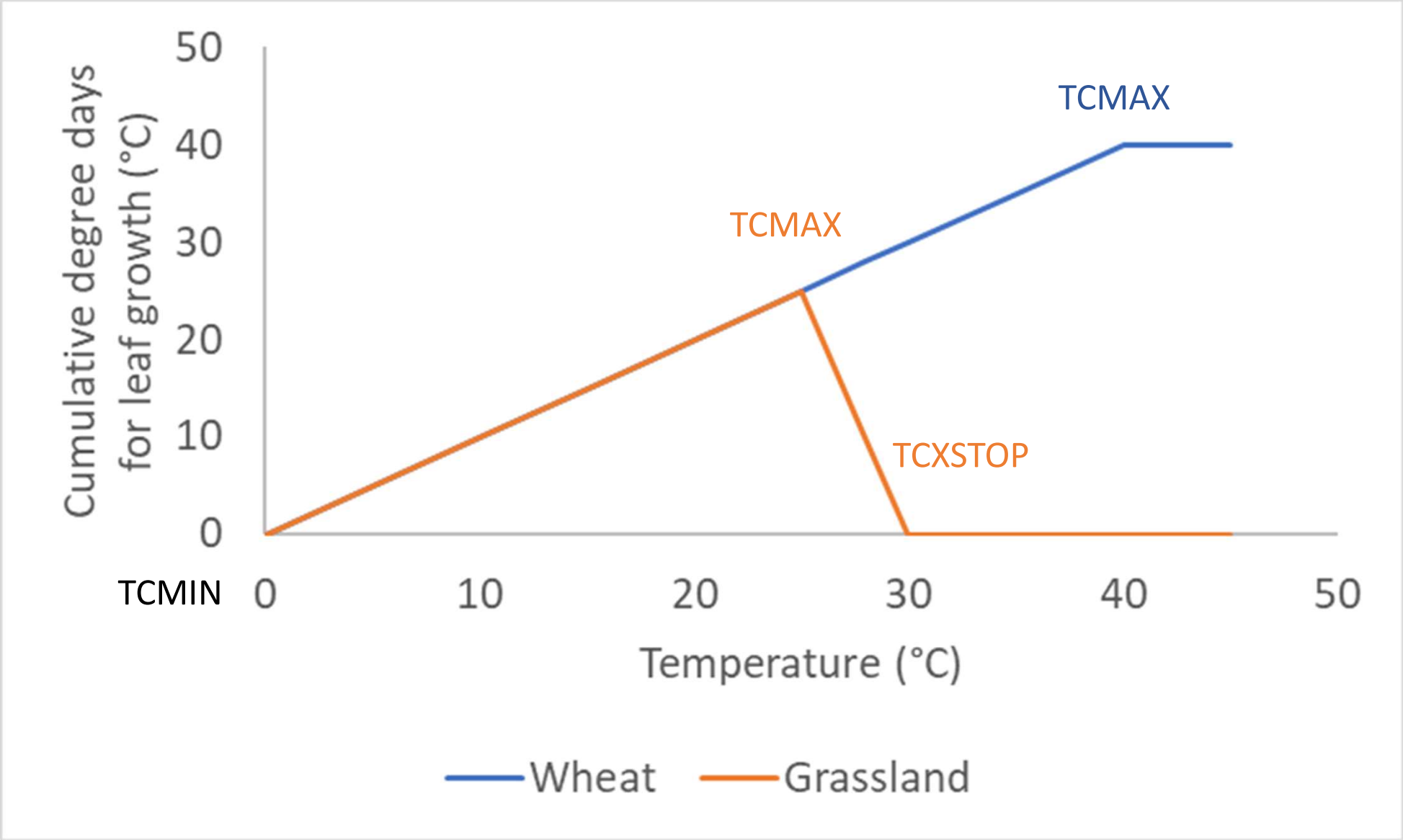
Effect of temperature

$$DELTAI_{stress}(I) = \min\left(TURFAC(I), INNLAI(I)\right) \cdot EXOLAI(I)$$

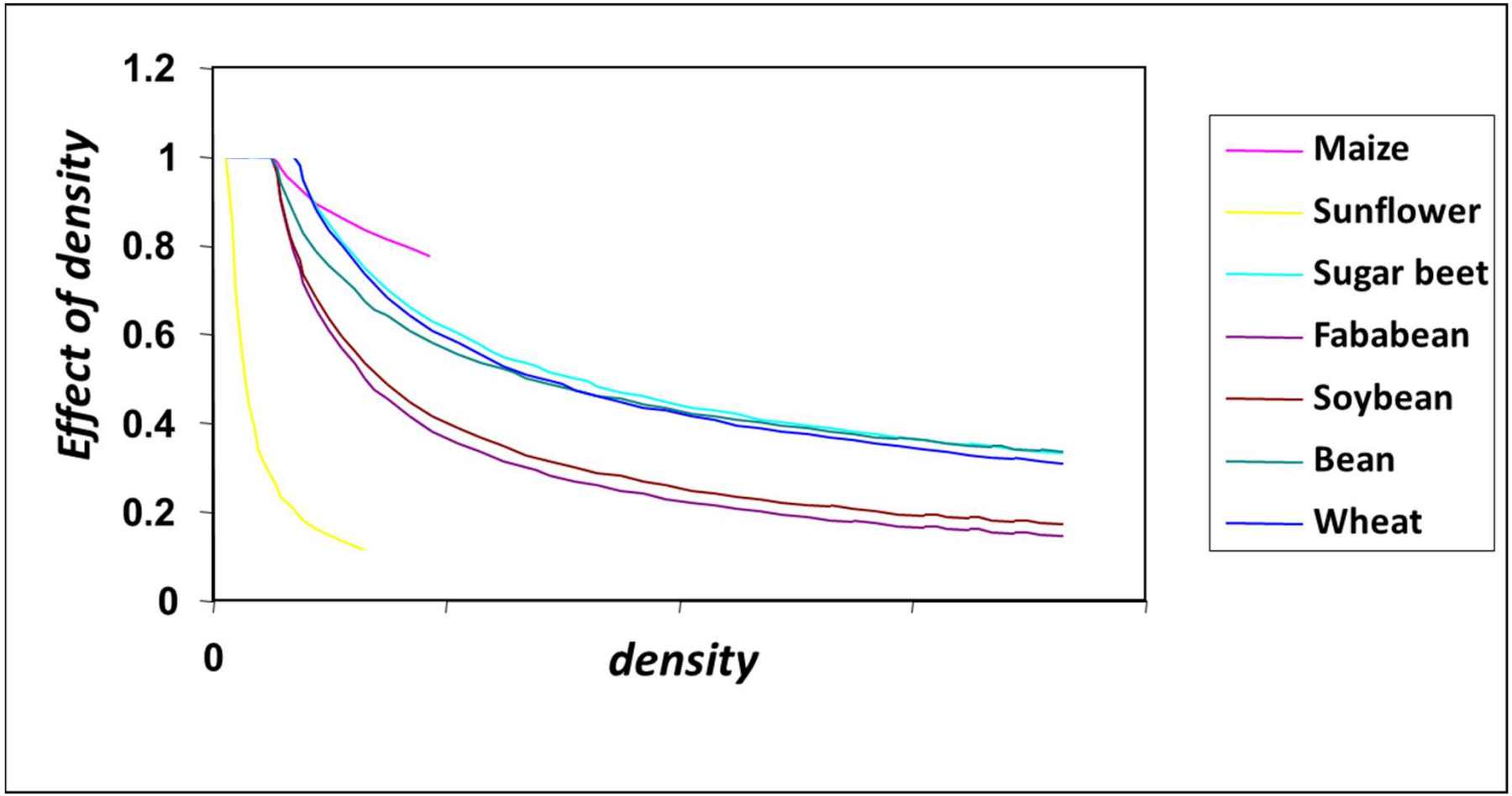
Leaf area index is a result of leaf growth and leaf senescence.



Leaf area index: effect of temperature



Leaf area index: effect of plant density



Leaf area index: trophic stress

Determinate crops

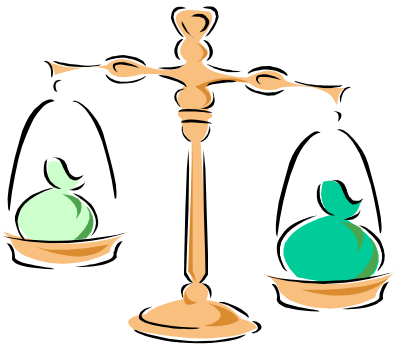
Trophic stress if radiations are limiting

Indeterminate crops

Sources :

Photosynthesis

Remobilizations



Sources/Sinks ratio

Sinks :

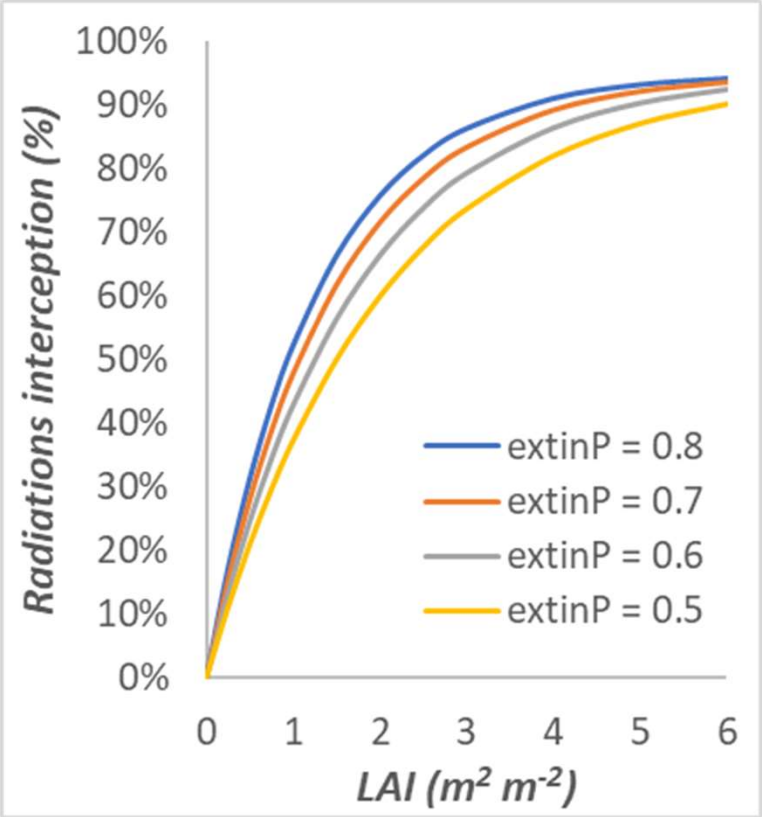
Fruits potential growth

Vegetative growth

Radiation interception

Beer law

For homogeneous soil cover



Radiative transfer

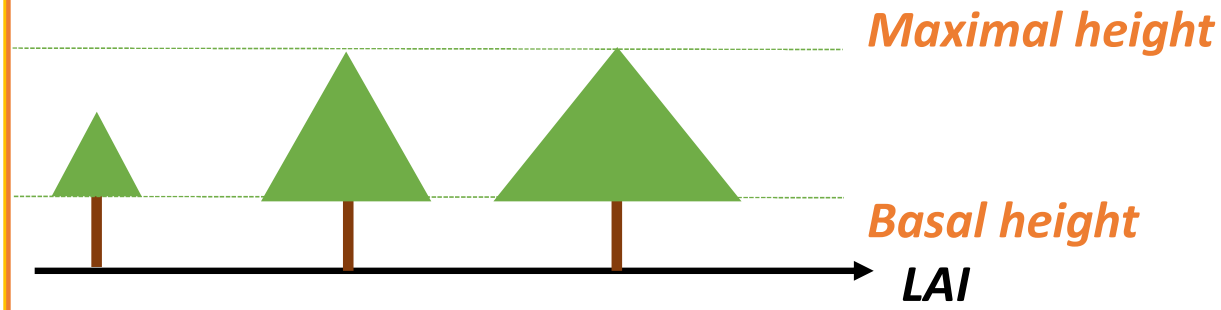
For raw crop (vineyard) or intercrop

1. ***Choose a shape***

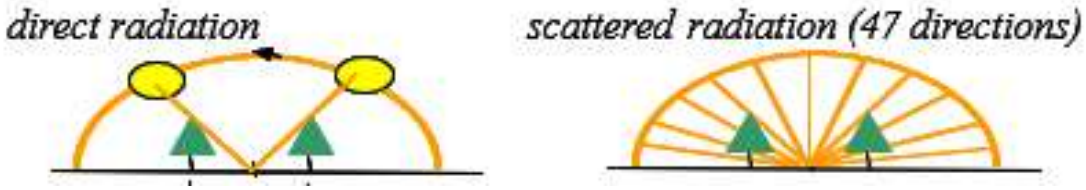
Leaf density profile type

Width / thickness ratio constant

2. ***Shape growth = f(LAI)***

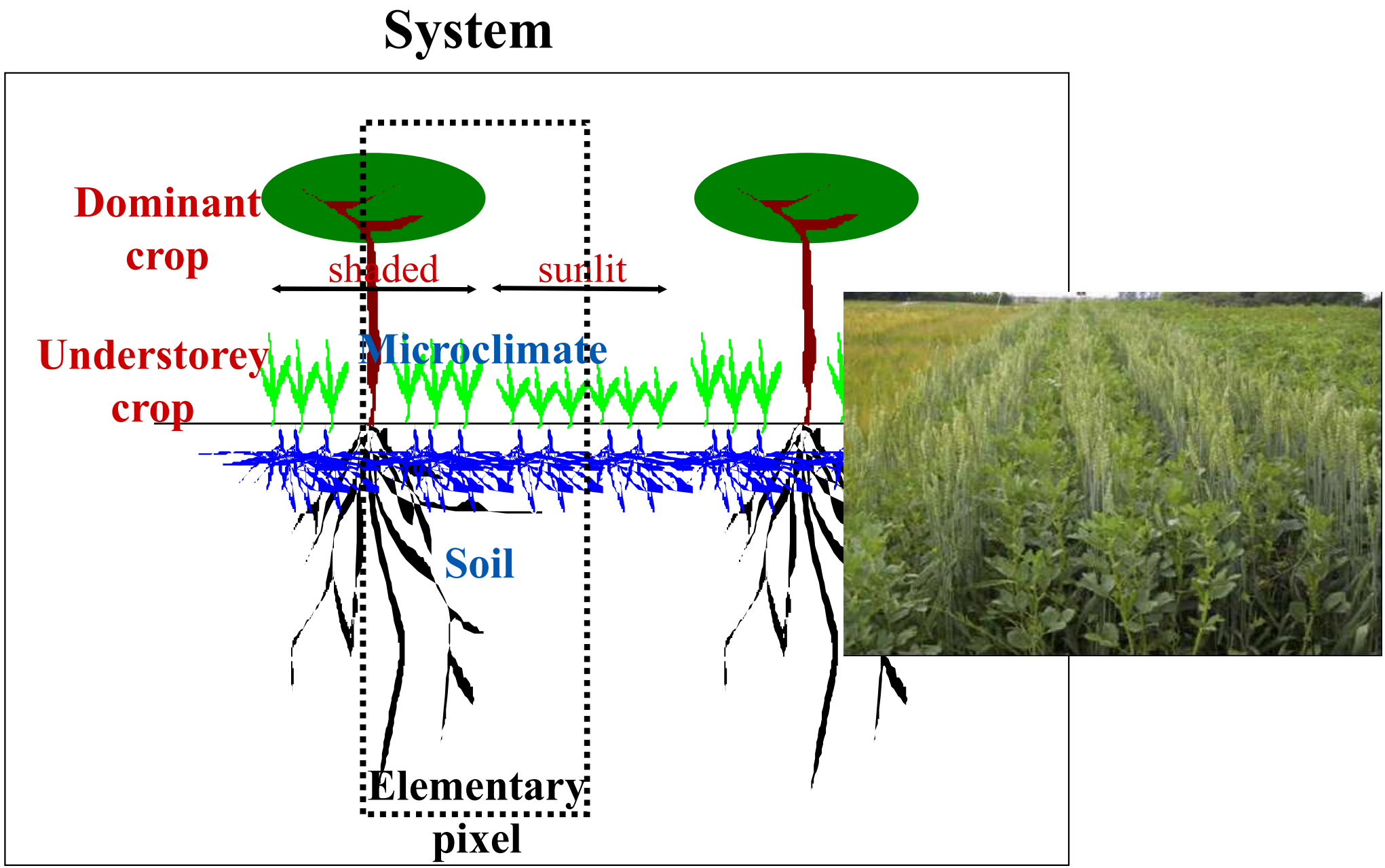


3. ***Radiations transfer***



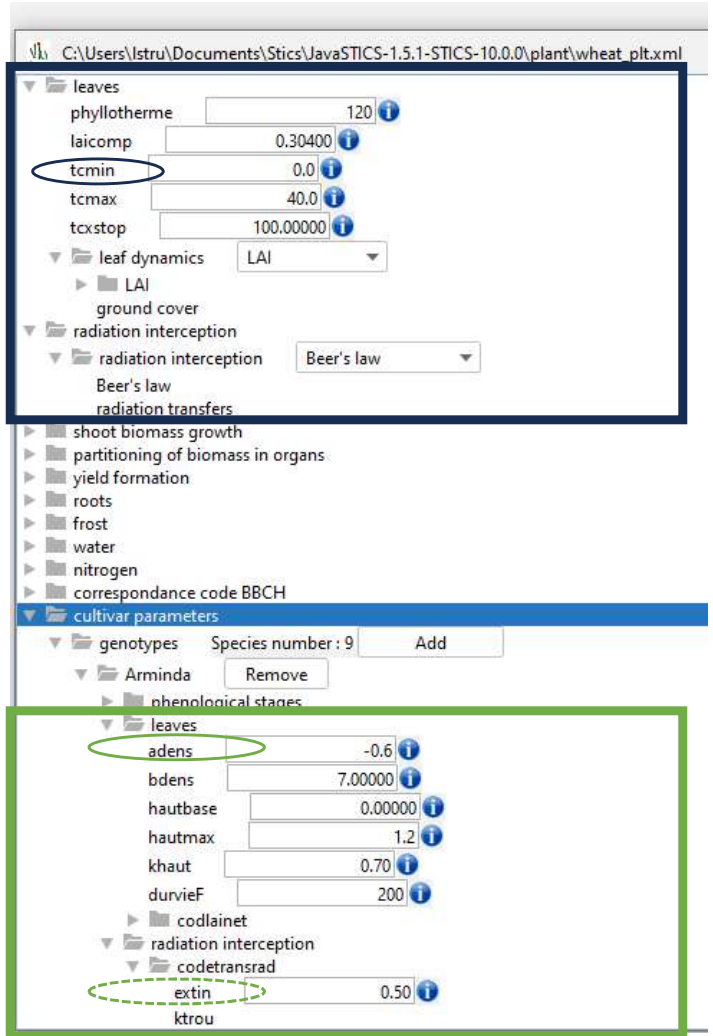
Daily simulation of transmitted and intercepted radiations

Intercropping



Leaf area index and radiation interception

- Plant parameters and options for simulation of LAI growth and radiations interception are available in the plant file



Leaf area index and radiation interception

- Plant parameters and options for simulation of LAI growth and radiations interception are available in the plant file
- LAI growth takes into account the effect of developmental stage, temperature, sowing density, abiotic stresses (N stress, water stress and flooding) and trophic competition (for indeterminate crops)
- There is 2 options to simulate radiation interception:
  - ✓ Beer law: for homogeneous ground cover
  - ✓ Radiative transfert: for raw crops or inter-crop

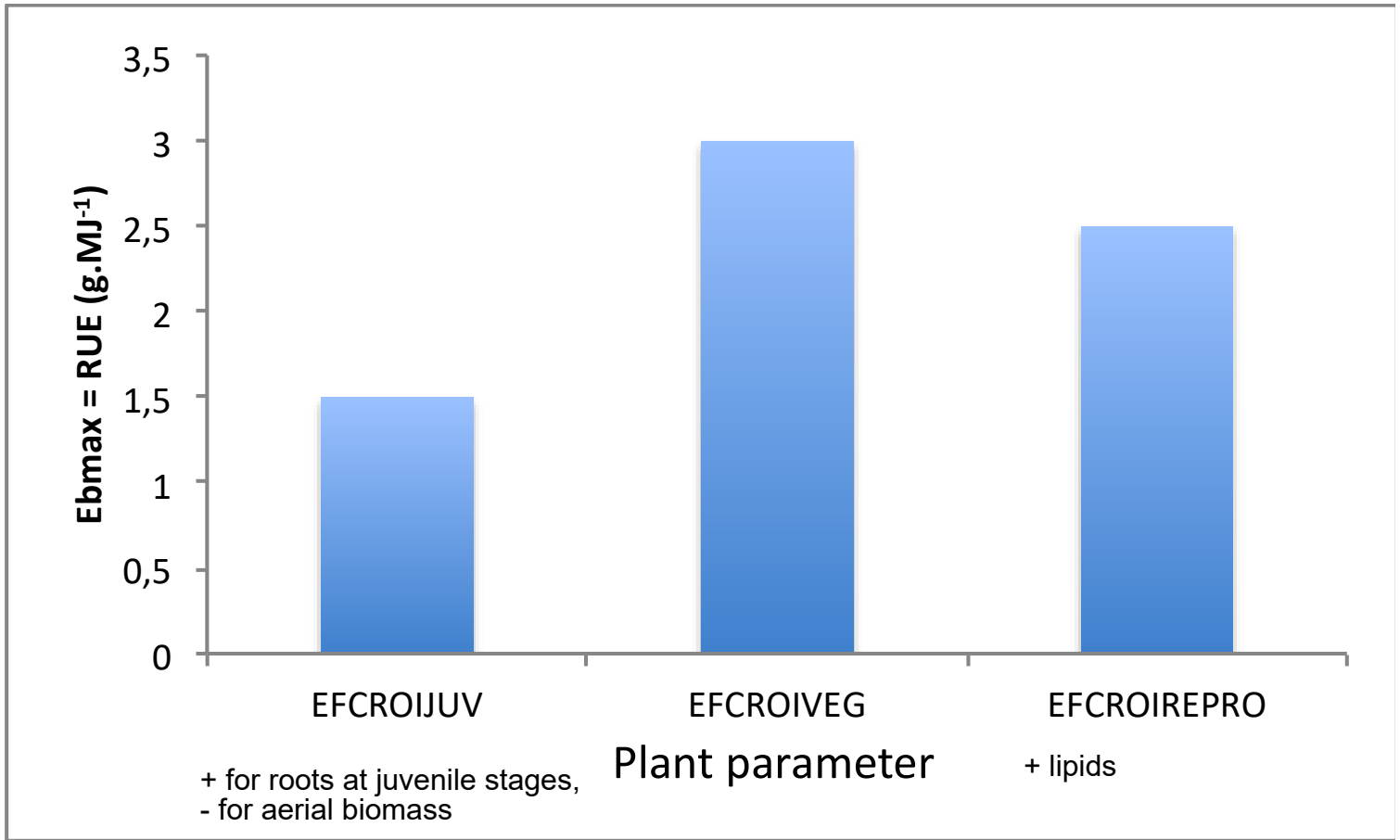
- Do you have any question ?

Biomass production

Biomass production

Daily potential biomass production

$$DLTAMS(I) = \left[EBMAX(I) \cdot RAINT(I) - COEFB_G \cdot RAINT(I)^2 \right] \cdot FTEMP(I) \cdot SWFAC(I-1) \cdot INNS(I-1) \cdot EXOBIOM(I-1) \cdot FCO2 + DLTAREMOBIL(I-1)$$



Biomass production

Daily potential biomass production

$$DLTAMS(I) = \left[EBMAX(I) \cdot RAINT(I) - COEFB_G \cdot RAINT(I)^2 \right] \cdot FTEMP(I) \cdot SWFAC(I-1) \cdot INNS(I-1) \cdot EXOBIOM(I-1) \cdot FCO2 + DLTAREMOBIL(I-1)$$

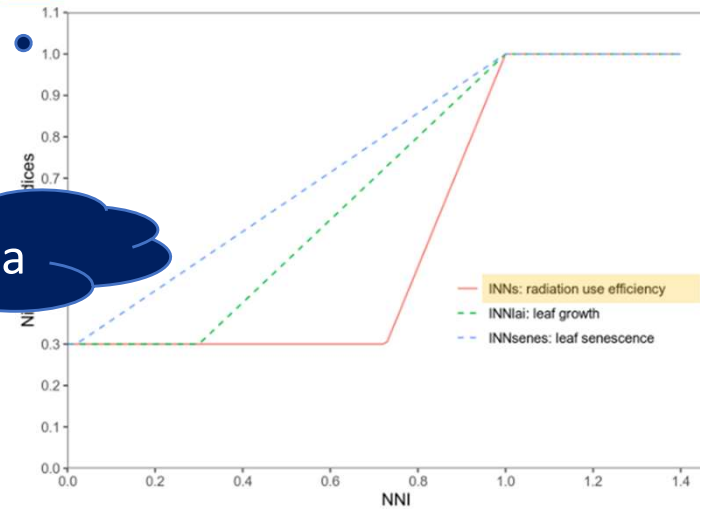
Effect of radiation saturation

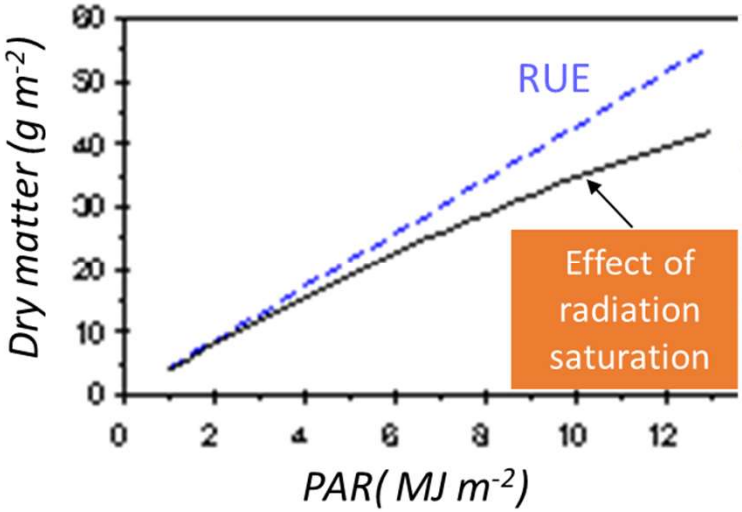
T°C effect

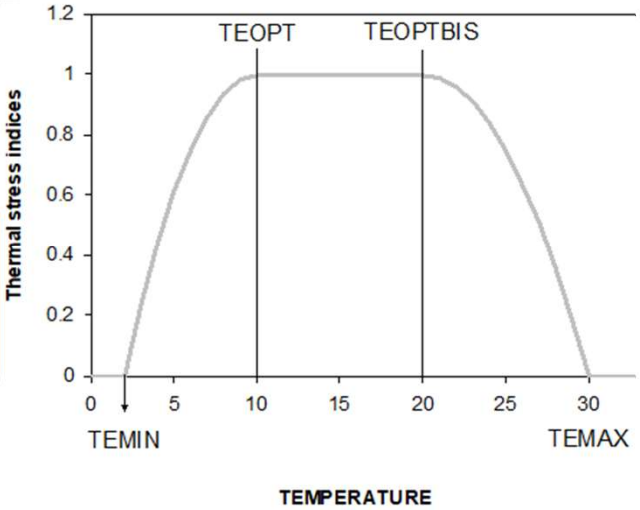
Effect of water stress

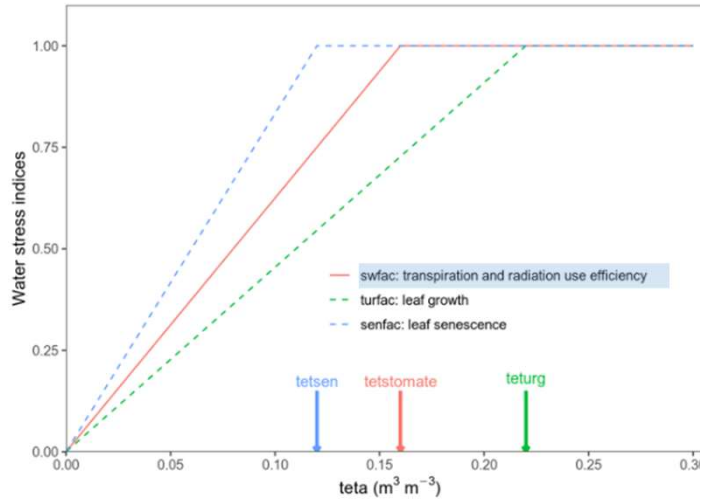
Effect of N stress

Effect of anoxia









Biomass production

Daily potential biomass production

C3 or C4

$$DLTAMS(I) = \left[EBMAX(I) \cdot RAINT(I) - COEFB_G \cdot RAINT(I)^2 \right] \cdot FTEMP(I) \cdot SWFAC(I-1) \cdot INNS(I-1) \cdot EXOBIOM(I-1) \cdot FCO2 + DLTAREMOBIL(I-1)$$

Effect of radiation saturation

T°C effect

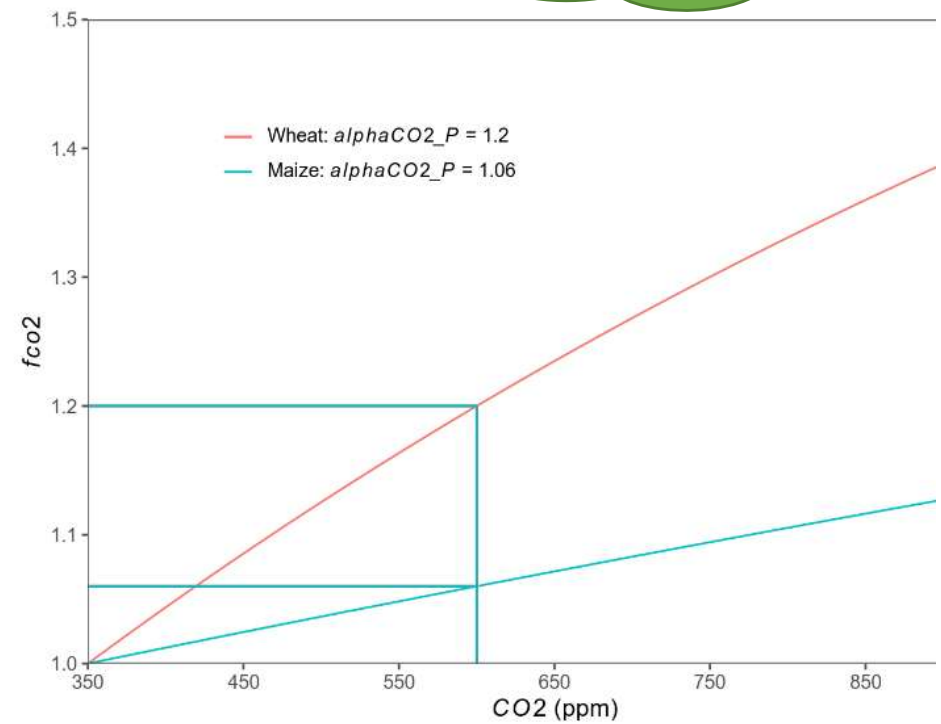
Effect of water stress

Effect of N stress

Effect of anoxia

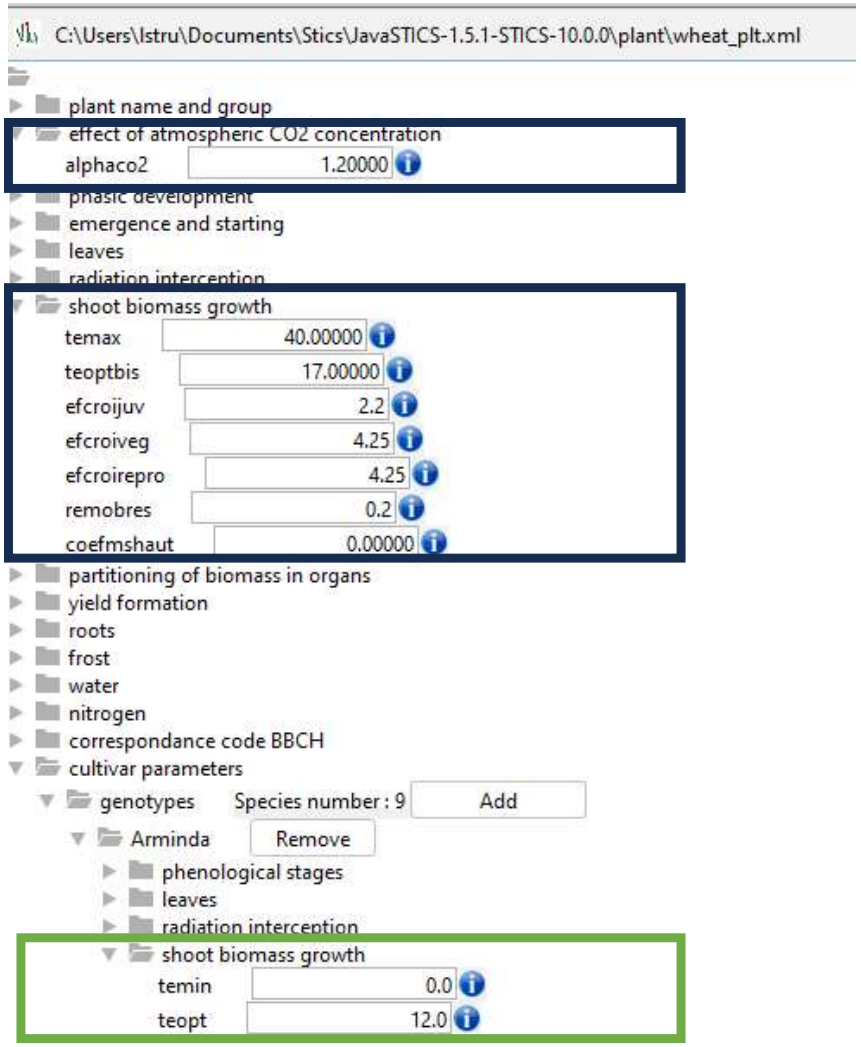
Effect of CO_2

Reserves remobilisation



Biomass production

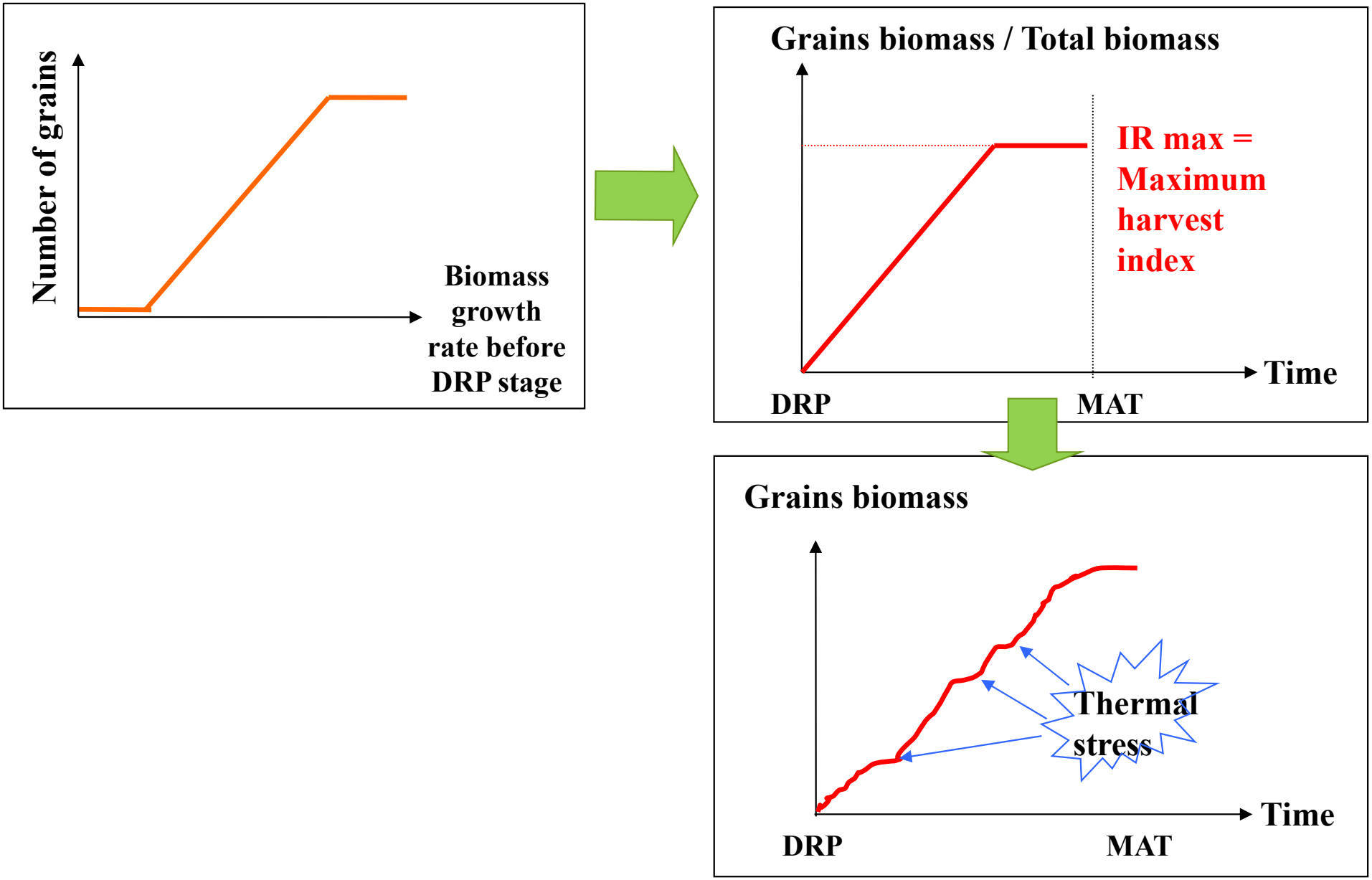
- Plant parameters for biomass production are available in the plant file:



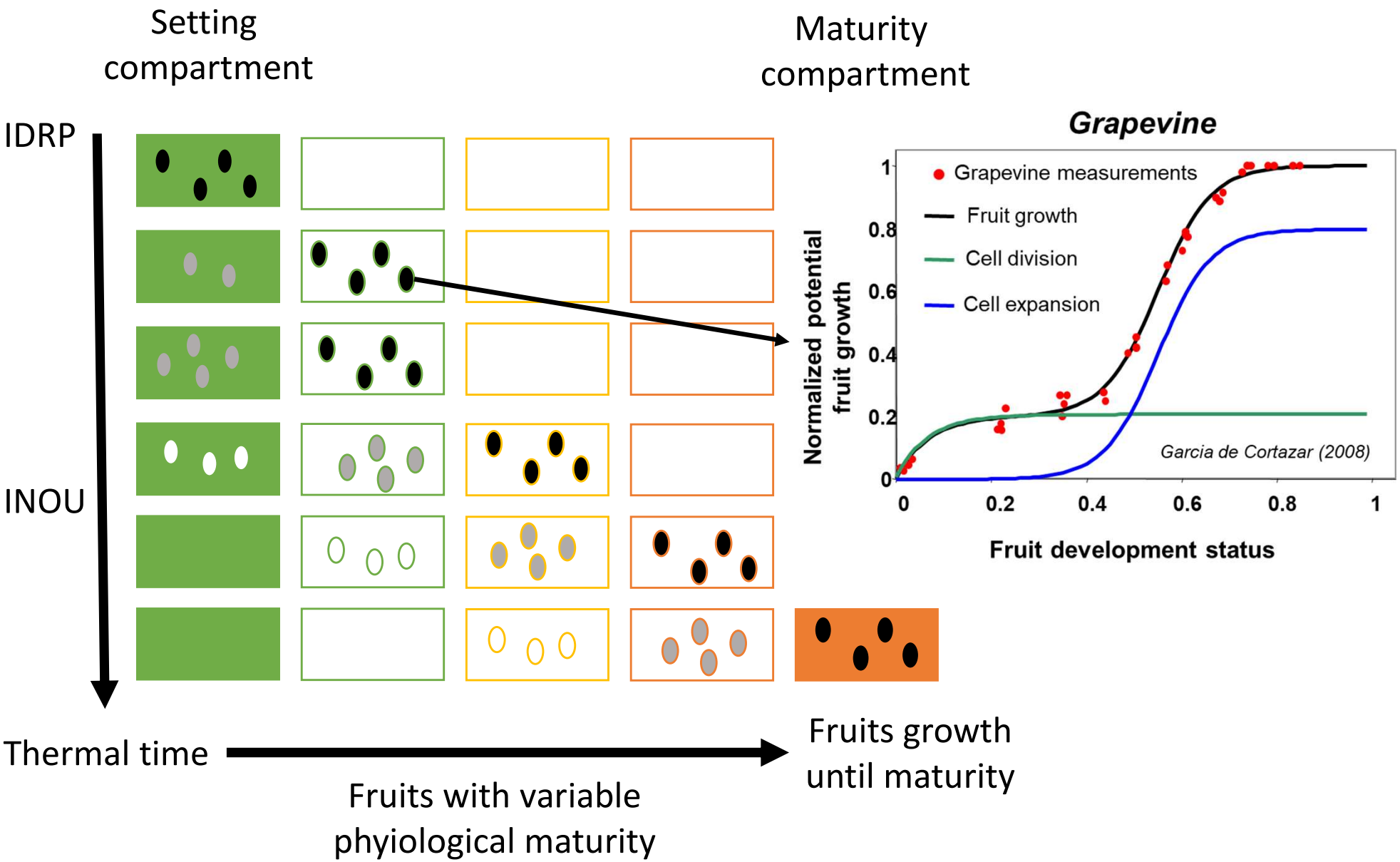
- Do you have any question ?

Yield

Yield formation for determinate growing crops

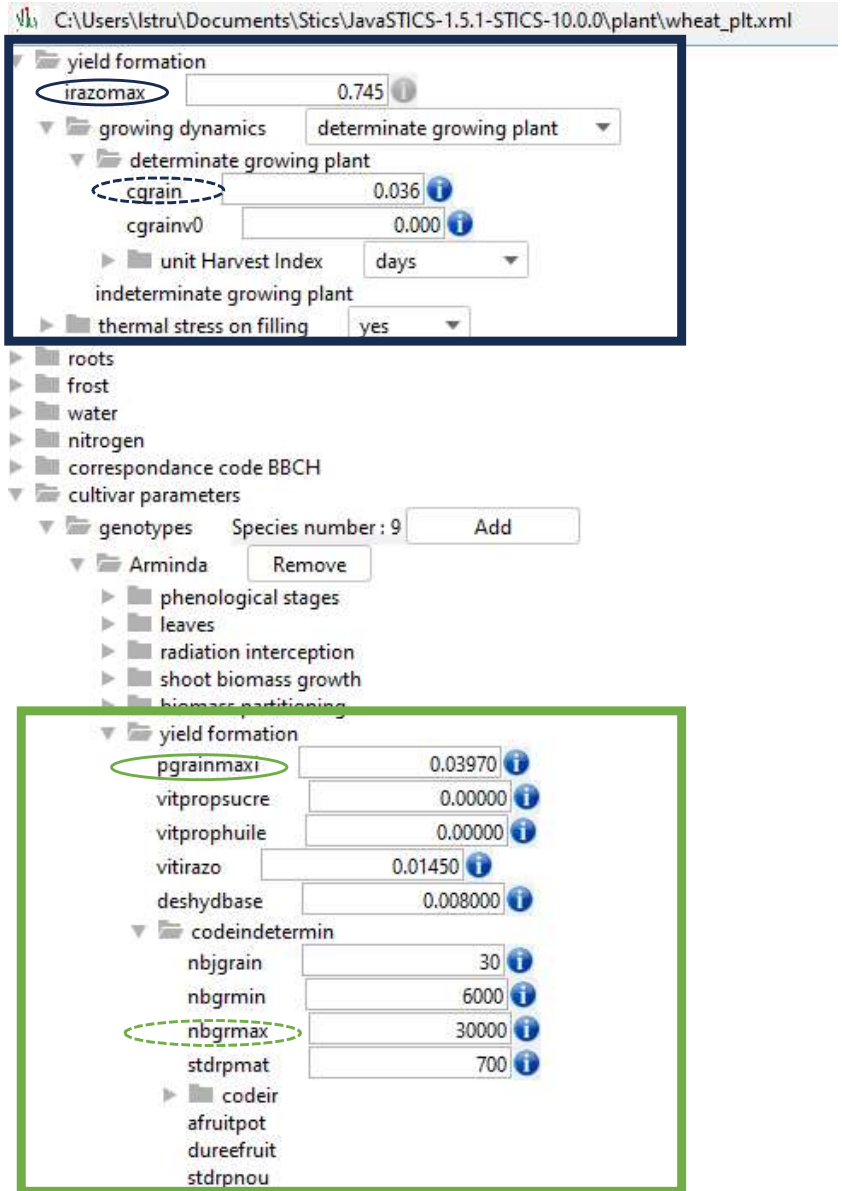


Yield formation for indeterminate growing crops



Yield

- Plant parameters for fruits production are available in the plant file:



FIN

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